

A Brief Course in Variational Methods and Analytical Mechanics

From the brachistochrone to Hamilton's equations

Lecture notes

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Abstract

This document develops Lagrangian and Hamiltonian mechanics from the calculus of variations, with every step of the main logical chain displayed. Part I opens with two problems that ordinary calculus cannot pose — the brachistochrone and the catenary — builds the variational framework rigorously (functionals, variations, the Euler–Lagrange equation, first integrals, constrained problems), solves both opening problems in full, and then turns to mechanics: Hamilton's principle, generalized coordinates and constraints, conservation laws and Noether's theorem, the pendulum, and the double pendulum at small angles, whose normal modes require and receive the general theory of the generalized eigenvalue problem. Part II introduces Hamiltonian mechanics: the Legendre transform and Hamilton's equations (derived twice); then, as the stress test of the formalism, the pendulum and the double pendulum are solved a second time, completely, in Hamiltonian form — the exact pendulum reduced to its elliptic quadrature, the double pendulum's exact Hamiltonian constructed, its linearized flow decoupled by mode variables, and its general small-angle solution given in closed form — followed by Poisson brackets and phase-space structure including Liouville's theorem. Physical motivation is kept in view throughout: each formal step is preceded by the question it answers. Proofs whose details are purely technical are placed in appendices; every statement used in the main text is either proved there or explicitly pointed to its appendix. The main text keeps all key steps, all logic, and all physics.

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1 Conventions and Notation

Dots denote time derivatives; primes denote derivatives with respect to the spatial variable of a curve, $y' = dy/dx$. Vectors carry arrows ($\vec{q}$, $\vec{p}$), matrices are sans-serif ($\mathbf{M}$, $\mathbf{K}$). Indices i, j run over generalized coordinates, $1, \dots, n$; the summation convention is *not* used — sums are written out.

Convention 1.1 (Functionals). A *functional* is a map that eats a whole function and returns a number; we write square brackets, $J[y]$, to distinguish it from a function $f(x)$ of finitely many variables. The domain will always be a space of sufficiently smooth functions with prescribed boundary behavior, stated when it matters:

$$\mathcal{C} = \{ y \in C^2([a, b]) : y(a) = y_A, y(b) = y_B \}. \quad (1)$$

Convention 1.2 (Variations, and what δ means). Physics texts write δy for “an infinitesimal variation of the function y .” We give this a precise meaning once, and then use the physicists’ notation freely. Fix $y \in \mathcal{C}$ and a *test function*

$$\eta \in C^2([a, b]) \quad \text{with} \quad \eta(a) = \eta(b) = 0 \quad (2)$$

(so that $y + \varepsilon\eta$ stays in $\mathcal{C}$ for every real ε : the endpoints are pinned). The *first variation* of J at y in the direction η is the ordinary derivative

$$\delta J[y; \eta] \equiv \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} J[y + \varepsilon\eta] \quad (3)$$

— a *Gateaux derivative*: one real parameter, ordinary calculus. The symbol δy is shorthand for $\varepsilon\eta$ at first order, and an equation like $\delta J = 0$ means: $\delta J[y; \eta] = 0$ for every admissible test function η . All manipulations with δ below are theorems about this definition, not a new calculus.

Convention 1.3 (Stationary, not minimal). A function y at which $\delta J[y; \eta] = 0$ for all η is a *stationary point* (or *extremal*) of J . Whether it is a minimum, maximum, or saddle is a separate question, governed by the second variation, and largely irrelevant to mechanics: the physical principle below demands stationarity only. The traditional name “principle of least action” overstates; we will say *stationary action*.

Citations across the series. References to the companion documents use the bracketed register [Osc. § n] (the oscillator reference) and [QM. § n] (the quantum mechanics notes), always against the cited document’s current numbering.

Part I

Variational Calculus and Lagrangian Mechanics

2 The Calculus of Variations

2.1 Two problems that ordinary calculus cannot pose

Ordinary calculus optimizes over *numbers*: find x minimizing $f(x)$. The following two classical problems ask for an optimum over *curves* — infinitely many degrees of freedom — and launched the subject.

Problem 1: the brachistochrone (Johann Bernoulli, 1696). A bead slides without friction along a wire from point A to a lower point B , released from rest, under uniform gravity. *Along which shape of wire is the travel time least?* Set A at the origin, measure y *downward*, and describe the wire by a function $y(x)$ with $y(0) = 0$, $y(x_B) = y_B$. Two ingredients convert the question into a formula:

- **Energy conservation** (the wire is frictionless and does no work on the bead — the constraint force is perpendicular to the motion): starting from rest, $\frac{1}{2}mv^2 = mgy$, so the speed depends only on the depth,

$$v = \sqrt{2gy}. \quad (4)$$

- **Arc length**: a step dx along the curve covers distance $ds = \sqrt{1 + y'^2} dx$.

Time is distance over speed, summed along the curve:

$$T[y] = \int_0^{x_B} \frac{ds}{v} = \int_0^{x_B} \sqrt{\frac{1 + y'(x)^2}{2gy(x)}} dx \quad (5)$$

The unknown is the *function* y . Note the genuine tension in the problem: a steep start builds speed early but lengthens the path; a straight line is shortest but slow at the start. Neither extreme wins (Fig. 1), and no finite list of parameters captures “all curves.”

Problem 2: the catenary (Galileo’s wrong guess, resolved 1691). A chain of fixed length L and uniform mass per length μ hangs between two pegs. *What shape does it assume?* The physics: the chain settles into the shape of *least potential energy* compatible with its fixed length. With height $y(x)$ (now measured upward) and the same arc length element,

$$U[y] = \mu g \int_a^b y \sqrt{1 + y'^2} dx \quad \text{to be minimized subject to} \quad \Lambda[y] = \int_a^b \sqrt{1 + y'^2} dx = L \quad (6)$$

— an optimization over curves *with a constraint that is itself a functional*. (Galileo guessed a parabola; Fig. 2 shows how close and how wrong that is.)

A third problem, already solved by nature. Both problems have a famous older sibling: *Fermat's principle* (1662) — light traveling between two points takes the path of least *time*. In a medium with position-dependent refractive index n , light moves at speed c/n , so the principle reads: minimize $\int n \, ds/c$ — again a functional of the path. Every law of geometric optics (straight rays, reflection, Snell's law of refraction) is a consequence, and this is the prototype of a lesson physics has since learned repeatedly: *a law of nature can be packaged not as a rule applied instant by instant, but as an optimization over entire histories*. Part of this document's business is to show that all of mechanics admits exactly such a packaging.

Meanwhile, the structure shared by all three problems is one and the same: extremize $J[y] = \int F(x, y, y') \, dx$ over curves with pinned endpoints. We now build the tool that solves them all — and then discover that mechanics itself has been secretly posed in this form all along.

2.2 The Euler–Lagrange equation

Theorem 2.1 (Euler–Lagrange). *Let $F(x, u, w)$ be C^2 in its three arguments and let*

$$J[y] = \int_a^b F(x, y(x), y'(x)) \, dx \quad (7)$$

on the class $\mathcal{C}$ of Convention 1.1. If $y \in \mathcal{C}$ is stationary — $\delta J[y; \eta] = 0$ for all test functions η — then y satisfies

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \quad \text{for all } x \in [a, b].} \quad (8)$$

Here $\partial F/\partial y$ and $\partial F/\partial y'$ denote partial derivatives of $F(x, u, w)$ with respect to its second and third slots, evaluated along the curve.

Proof. The proof is four moves; each is displayed.

Move 1: reduce to one variable. Fix a test function η and consider the one-parameter family of competitor curves $y + \varepsilon\eta$ (each in $\mathcal{C}$: the pinned endpoints are undisturbed since $\eta(a) = \eta(b) = 0$). Along this family the functional becomes an *ordinary function of one real variable*,

$$\Phi(\varepsilon) \equiv J[y + \varepsilon\eta] = \int_a^b F(x, y(x) + \varepsilon\eta(x), y'(x) + \varepsilon\eta'(x)) \, dx, \quad (9)$$

and the hypothesis $\delta J[y; \eta] = 0$ is, by Convention 1.2, precisely the statement $\Phi'(0) = 0$. The infinite-dimensional problem has been probed along a line; all of ordinary calculus now applies to Φ .

Move 2: differentiate under the integral. Two things must be done: compute the ε -derivative of the integrand, and justify pulling $d/d\varepsilon$ inside the integral. For the first, fix x and apply the chain rule to $\varepsilon \mapsto F(x, u(\varepsilon), w(\varepsilon))$ with $u = y + \varepsilon\eta$, $w = y' + \varepsilon\eta'$ (so $du/d\varepsilon = \eta(x)$, $dw/d\varepsilon = \eta'(x)$):

$$\frac{\partial}{\partial \varepsilon} F(x, y + \varepsilon\eta, y' + \varepsilon\eta') = \frac{\partial F}{\partial u}(x, y + \varepsilon\eta, y' + \varepsilon\eta') \eta(x) + \frac{\partial F}{\partial w}(x, y + \varepsilon\eta, y' + \varepsilon\eta') \eta'(x). \quad (10)$$

For the second: F is C^2 and y, η are C^2 , so the right side of (10) is continuous in (x, ε) on the compact rectangle $[a, b] \times [-1, 1]$; this is exactly the hypothesis of the Leibniz rule (stated and proved as Appendix A), which licenses

$$\Phi'(\varepsilon) = \int_a^b \frac{\partial}{\partial \varepsilon} F(x, y + \varepsilon\eta, y' + \varepsilon\eta') \, dx. \quad (11)$$

Now set $\varepsilon = 0$: the partial derivatives in (10) are then evaluated at $(x, y(x), y'(x))$ — i.e. *along the original curve* — and we adopt the standing shorthand $\partial F/\partial y, \partial F/\partial y'$ for exactly these evaluated slot-derivatives (as announced in the theorem statement):

$$\Phi'(0) = \int_a^b \left[\frac{\partial F}{\partial y} \eta(x) + \frac{\partial F}{\partial y'} \eta'(x) \right] dx. \quad (12)$$

This quantity — linear in the probe η — is the *first variation* of J ; the notation $\delta J = \int(\dots) \delta y dx$ of physics texts is (12) with $\delta y = \varepsilon \eta$.

Move 3: integrate by parts to strip the derivative off η . The two terms of (12) probe the curve differently — one through η , one through η' — and cannot be compared until they speak the same language. The conversion is the product rule read backwards: along the curve, the function $x \mapsto \frac{\partial F}{\partial y'} \eta$ is C^1 (here the hypotheses do their work: $y \in C^2$ and $F \in C^2$ make $x \mapsto \partial F/\partial y'$ a C^1 function — its derivative is the chain-rule expansion (16) below, every term of which is continuous), and

$$\frac{d}{dx} \left(\frac{\partial F}{\partial y'} \eta \right) = \left[\frac{d}{dx} \frac{\partial F}{\partial y'} \right] \eta + \frac{\partial F}{\partial y} \eta'. \quad (13)$$

Integrate both sides over $[a, b]$; the left side is handled by the fundamental theorem of calculus, and rearranging,

$$\begin{aligned} \int_a^b \frac{\partial F}{\partial y'} \eta' dx &= \underbrace{\left[\frac{\partial F}{\partial y'} \eta \right]_a^b}_{=0 \text{ since } \eta(a)=\eta(b)=0} - \int_a^b \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \eta dx. \end{aligned} \quad (14)$$

The boundary term dies *because the endpoints are pinned* — this is where the boundary conditions enter the theory, and the only place. Substituting,

$$0 = \Phi'(0) = \int_a^b \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta(x) dx \quad \text{for every test function } \eta. \quad (15)$$

Move 4: localize. From “the integral against every test function vanishes” we want “the bracket vanishes pointwise.” That step is the

Fundamental Lemma of the Calculus of Variations. *If g is continuous on $[a, b]$ and $\int_a^b g(x) \eta(x) dx = 0$ for every $\eta \in C^2$ vanishing at the endpoints, then $g \equiv 0$.*

The logic of its proof is worth carrying in the main text even though the construction lives in Appendix B: if g were nonzero somewhere, say $g(x_0) > 0$, then by continuity $g > 0$ on a small interval around x_0 ; choosing η to be a smooth *bump* — positive inside that interval, zero outside — makes the integral strictly positive, a contradiction. Test functions are probes; enough probes pin g down everywhere. The bracket in (15) is continuous (exactly what Move 3 checked), so the lemma applies to it, and the conclusion of the theorem is reached: for every $x \in [a, b]$,

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

□

Remark 2.2 (What kind of object (8) is). The Euler–Lagrange equation converts a problem over an infinite-dimensional space of curves into an ordinary differential equation — generally second order, since $\frac{d}{dx}(\partial F/\partial y')$ produces a y'' through the chain rule:

$$\frac{d}{dx} \frac{\partial F}{\partial y'} = \frac{\partial^2 F}{\partial x \partial y'} + \frac{\partial^2 F}{\partial y \partial y'} y' + \frac{\partial^2 F}{\partial y'^2} y''. \quad (16)$$

Stationarity is a *local, differential* condition on the curve — this is the first deep lesson of the subject: a globally posed optimization is answered by a pointwise law. It is also a *necessary* condition only; Convention 1.3 applies.

Remark 2.3 (Several unknown functions). If the functional depends on n functions, $J[\vec{q}] = \int_a^b F(x, q_1, \dots, q_n, q'_1, \dots, q'_n) dx$, run the same proof with independent test functions η_i attached to each q_i : choosing all but one η_i zero isolates one equation at a time, giving the *system*

$$\boxed{\frac{\partial F}{\partial q_i} - \frac{d}{dx} \frac{\partial F}{\partial q'_i} = 0, \quad i = 1, \dots, n.} \quad (17)$$

Nothing else changes. Mechanics will use exactly this form, with $x \rightarrow t$.

Example 2.4 (Calibration: the shortest path is a straight line). Before the hard problems, the sanity check. The length of a graph $y(x)$ between two points is $J[y] = \int_a^b \sqrt{1 + y'^2} dx$, so $F = \sqrt{1 + y'^2}$ with

$$\frac{\partial F}{\partial y} = 0, \quad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}}. \quad (18)$$

The Euler–Lagrange equation (8) says $\frac{d}{dx} (y' / \sqrt{1 + y'^2}) = 0$, i.e. $y' / \sqrt{1 + y'^2} = c$ constant; solving, $y'^2 = c^2 / (1 - c^2)$ is constant, so y' is constant: a straight line, through the two pinned endpoints. The machinery returns the right answer on the one problem where we already knew it — the same calibration discipline as checking a resonance formula at $\omega \rightarrow 0$.

With the equation in hand, the section returns to its two opening problems. Both opening functionals share one feature the general theory rewards: their integrands carry *no explicit x -dependence*. The next subsection converts that feature into a ready-made first integration, and the two subsections after it collect the brachistochrone and the catenary in full — the promised payoff of the machinery just built.

2.3 First integrals

Two structural shortcuts, both indispensable, both provable in a few lines.

Proposition 2.5 (Cyclic variable). *If F does not depend explicitly on y (i.e. $\partial F / \partial y \equiv 0$), then along any extremal*

$$\frac{\partial F}{\partial y'} = \text{const.} \quad (19)$$

Proof. Equation (8) collapses to $\frac{d}{dx} (\partial F / \partial y') = 0$. □

Proposition 2.6 (Beltrami identity). *If F does not depend explicitly on x (i.e. $\partial F / \partial x \equiv 0$), then along any extremal*

$$\boxed{F - y' \frac{\partial F}{\partial y'} = \text{const.}} \quad (20)$$

Proof. Differentiate the left side along the curve and watch everything cancel. By the chain rule (no explicit x -dependence),

$$\frac{dF}{dx} = \frac{\partial F}{\partial y} y' + \frac{\partial F}{\partial y'} y'', \quad (21)$$

while

$$\frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} \right) = y'' \frac{\partial F}{\partial y'} + y' \frac{d}{dx} \frac{\partial F}{\partial y'}. \quad (22)$$

Subtracting,

$$\frac{d}{dx} \left(F - y' \frac{\partial F}{\partial y'} \right) = y' \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right] = y' \cdot 0 = 0 \quad (23)$$

by the Euler–Lagrange equation. (This identity will return in §3.3 in mechanical form: it *is* energy conservation.) $\square$

The practical payoff: both opening problems have integrands with no explicit x -dependence, so both are governed by the *first-order* equation (20) instead of the second-order (8) — one integration is handed to us for free.

2.4 Solution of the brachistochrone

The problem that opened the section can now be solved. Apply Beltrami to (5). Drop the constant $1/\sqrt{2g}$ and take

$$F(y, y') = \sqrt{\frac{1 + y'^2}{y}}, \quad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{y} \sqrt{1 + y'^2}}. \quad (24)$$

Then

$$F - y' \frac{\partial F}{\partial y'} = \frac{1 + y'^2}{\sqrt{y} \sqrt{1 + y'^2}} - \frac{y'^2}{\sqrt{y} \sqrt{1 + y'^2}} = \frac{1}{\sqrt{y} (1 + y'^2)} = \text{const}, \quad (25)$$

i.e.

$$y (1 + y'^2) = 2a \quad (2a \text{ a positive constant}). \quad (26)$$

Solve by a substitution tailored to the structure: since $y \leq 2a$ along the curve, write

$$y = a (1 - \cos \theta) = 2a \sin^2 \frac{\theta}{2}, \quad \theta \in [0, \pi], \quad (27)$$

with θ a parameter along the curve. From (26),

$$y'^2 = \frac{2a - y}{y} = \frac{2a \cos^2(\theta/2)}{2a \sin^2(\theta/2)} \implies y' = \cot \frac{\theta}{2} \quad (28)$$

(positive branch: the curve initially descends with our downward- y convention). Then, using $dy = a \sin \theta d\theta$,

$$dx = \frac{dy}{y'} = a \sin \theta \tan \frac{\theta}{2} d\theta = 2a \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cdot \frac{\sin(\theta/2)}{\cos(\theta/2)} d\theta = 2a \sin^2 \frac{\theta}{2} d\theta = a(1 - \cos \theta) d\theta, \quad (29)$$

which integrates immediately (constant fixed by A at $\theta = 0$):

$$\boxed{x(\theta) = a(\theta - \sin \theta), \quad y(\theta) = a(1 - \cos \theta)} \quad (30)$$

— a *cycloid*: the curve traced by a point on the rim of a wheel of radius a rolling under the line $y = 0$. The two constants (a, θ_B) are fixed by requiring the curve to pass through B .

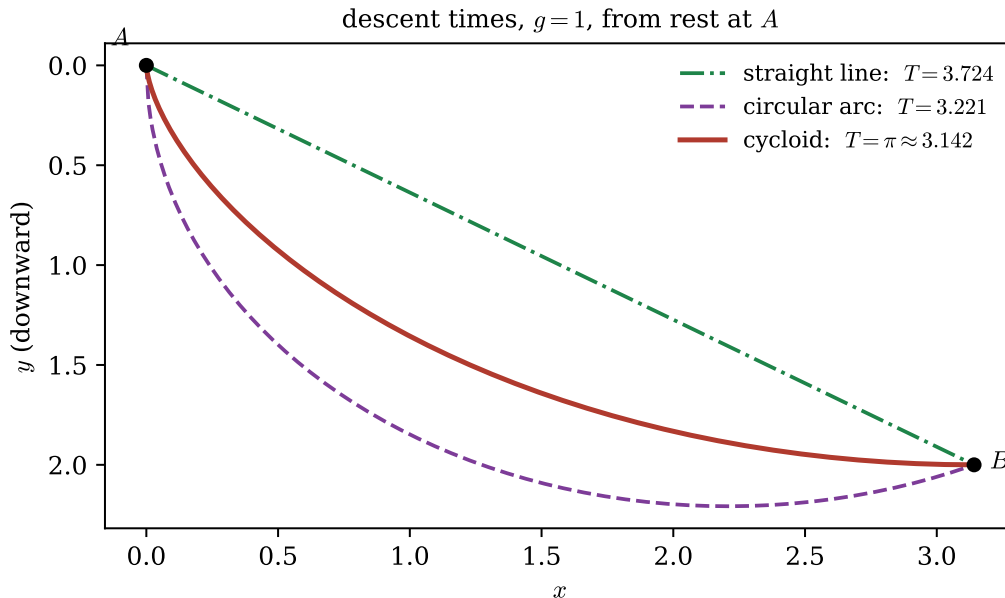


Figure 1: Three wires from A to $B = (\pi, 2)$ and their descent times (computed numerically from (5) with $g = 1$; the cycloid's time is exactly $\pi\sqrt{a/g}$). The straight line is shortest and slowest; the cycloid trades extra length for early speed and wins.

Remark 2.7 (Two bonuses of the parametrization). (i) Along the cycloid the travel-time element collapses beautifully: with $ds = 2a \sin(\theta/2) d\theta$ and $v = \sqrt{2gy} = 2\sqrt{ga} \sin(\theta/2)$,

$$dT = \frac{ds}{v} = \sqrt{\frac{a}{g}} d\theta \implies T = \sqrt{\frac{a}{g}} \theta_B : \quad (31)$$

the bead consumes *equal times for equal parameter intervals*. (ii) Releasing the bead from rest anywhere on the same cycloid, the time to reach the bottom is the same — the cycloid is also the *tautochrone*, the fact Huygens built into pendulum clocks. We state (ii) without proof; it is a short exercise from (i)'s machinery.

Remark 2.8 (Bernoulli's own solution: physics borrowed from optics). Johann Bernoulli solved the problem in 1697 without the Euler–Lagrange equation (which did not yet exist) by a physical translation: the bead's speed $v = \sqrt{2gy}$ depends only on depth, exactly like light in a stratified medium with index $n(y) \propto 1/v \propto 1/\sqrt{y}$. By Fermat's principle (§2.1), the least-time path of the bead must be the path such light would take — and Snell's law, $\sin \alpha/v = \text{const}$ layer by layer, is then a ready-made first integral. Writing $\sin \alpha = 1/\sqrt{1+y'^2}$ for a graph reproduces precisely our Beltrami result $y(1+y'^2) = \text{const}$. The calculus of variations is the common language in which the optical and mechanical statements are the same statement.

2.5 Constrained problems and the catenary

One opening problem remains, and it is the one the plain machinery cannot yet touch: the catenary (6) adds a new structural feature — the constraint is itself a functional (an *isoperimetric* constraint) — and closing it will add the last general-purpose tool of the section. The resolution is the infinite-dimensional version of Lagrange multipliers.

Theorem 2.9 (Isoperimetric multiplier rule). *Let $J[y] = \int_a^b F \, dx$ and $\Lambda[y] = \int_a^b G \, dx$ with F, G of class C^2 , and suppose y_* is stationary for J among curves in $\mathcal{C}$ satisfying $\Lambda[y] = L$. If y_* is not an unconstrained extremal of Λ itself, then there exists a constant $\lambda \in \mathbb{R}$ such that y_* satisfies the Euler–Lagrange equation of the combined integrand*

$$\tilde{F} = F + \lambda G. \quad (32)$$

The logic (full proof: Appendix D): with a constraint, one cannot vary in every direction η — only in directions that keep Λ fixed to first order. A two-parameter variation $y + \varepsilon_1 \eta_1 + \varepsilon_2 \eta_2$, with η_2 chosen to “repair” the constraint violation caused by η_1 , shows that δJ must be proportional to $\delta \Lambda$ direction by direction; the proportionality constant is λ . This is exactly the finite-dimensional picture — $\nabla f \parallel \nabla g$ on a constraint surface — promoted to function space. The hypothesis that y_* not extremize Λ is the nondegeneracy condition ($\nabla g \neq 0$) in disguise.

Solving the catenary. With $F = \mu g y \sqrt{1 + y'^2}$ and $G = \sqrt{1 + y'^2}$, absorb constants and set

$$\tilde{F} = (y + \lambda) \sqrt{1 + y'^2}, \quad (33)$$

(the multiplier shifts the height — already a hint of its physical meaning). No explicit x : Beltrami applies.

$$\frac{\partial \tilde{F}}{\partial y'} = \frac{(y + \lambda) y'}{\sqrt{1 + y'^2}}, \quad \tilde{F} - y' \frac{\partial \tilde{F}}{\partial y'} = (y + \lambda) \left[\sqrt{1 + y'^2} - \frac{y'^2}{\sqrt{1 + y'^2}} \right] = \frac{y + \lambda}{\sqrt{1 + y'^2}} = C. \quad (34)$$

Hence $(y + \lambda)^2 = C^2(1 + y'^2)$, i.e.

$$y' = \sqrt{\frac{(y + \lambda)^2}{C^2} - 1}. \quad (35)$$

Substitute $y + \lambda = C \cosh u$: then $dy = C \sinh u \, du$ and $y' = \sqrt{\cosh^2 u - 1} = \sinh u$, so

$$C \sinh u \frac{du}{dx} = \sinh u \implies \frac{du}{dx} = \frac{1}{C} \implies u = \frac{x - x_0}{C}, \quad (36)$$

$$\boxed{y(x) = C \cosh\left(\frac{x - x_0}{C}\right) - \lambda} \quad (37)$$

— the *catenary*. The three constants (C, x_0, λ) are fixed by the two endpoint conditions and the length constraint $\Lambda[y] = L$. Not a parabola: $\cosh$ and a parabola agree to quadratic order at the bottom and then part ways (Fig. 2), which is why Galileo’s guess survived casual inspection for half a century.

Remark 2.10 (The physical meaning of the multiplier). λ entered as bookkeeping and exits with a job: comparing with the statics of a hanging chain, $C = H/(\mu g)$ where H is the horizontal component of the chain’s tension (constant along the chain), and λ fixes the reference height at which the potential-plus-constraint total balances. Multipliers of constraints are *forces of constraint* in embryo — a theme §3.2 makes precise.

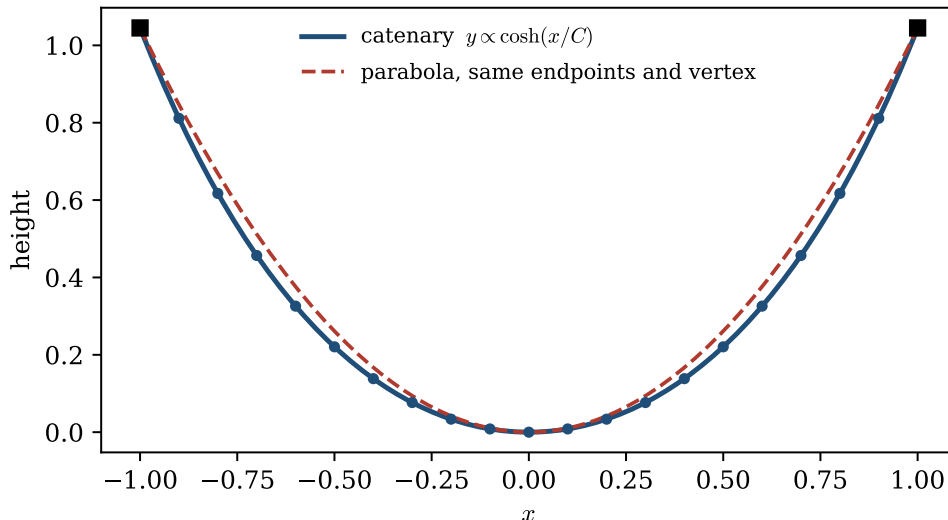


Figure 2: The catenary against the parabola with the same endpoints and vertex: close near the bottom (they share the quadratic term), visibly different toward the supports.

2.6 Covariance: why arbitrary coordinates are allowed

One more structural theorem before mechanics — the one that makes everything that follows practical.

Theorem 2.11 (Covariance of the Euler–Lagrange equations). *Let $Q_i = Q_i(q_1, \dots, q_n, t)$, $i = 1, \dots, n$, be a C^2 change of coordinates with C^2 inverse, and let $\tilde{L}$ be L rewritten in the new coordinates: $\tilde{L}(Q, \dot{Q}, t) = L(q(Q, t), \dot{q}(Q, \dot{Q}, t), t)$. Then a curve satisfies the Euler–Lagrange equations for L in the q ’s if and only if its image satisfies the Euler–Lagrange equations for $\tilde{L}$ in the Q ’s.*

Proof in Appendix C (two chain-rule identities and a cancellation). The content is one sentence: *the Euler–Lagrange equations do not care which coordinates you use*. Newton’s equations, by contrast, hold in Cartesian coordinates and mutate under coordinate change (centrifugal and Coriolis terms in polar coordinates, etc.). The variational formulation is coordinate-blind because the underlying object — “this curve makes $\int L dt$ stationary” — never mentions coordinates at all: stationarity is a property of the *curve*, and (8) is merely its expression in whichever chart is convenient. This theorem is the license behind the phrase *generalized coordinates*: angles, distances, differences of angles — anything that smoothly labels configurations is as good as x, y, z .

3 From Newton to Lagrange

Why rebuild mechanics, when $\vec{F} = m\vec{a}$ already works? Because Newton’s formulation, for all its power, has three practical irritations and hides one deep structure. The irritations: it is wedded to *vectors in Cartesian coordinates* (change to polar coordinates and fictitious terms sprout); it forces you to carry *constraint forces* — rod tensions, normal forces — as unknowns you must solve for and rarely want; and its symmetries are *invisible* — nothing in the notation $m\ddot{\vec{r}} = \vec{F}$ announces that momentum conservation is the same fact as the homogeneity of space. The variational reformulation answers all three at once: Theorem 2.11 makes any coordinates legal, the treatment of constraints

below makes constraint forces vanish before the calculation starts, and Noether's theorem (§3.3) turns symmetry into conservation by a two-line argument. The deep structure it exposes — one scalar function L generating all equations of motion, the dynamics as a stationarity condition on whole histories — is the form in which mechanics generalizes: field theory, relativity, and quantum mechanics are all written in this language, not in Newton's. The cost is one new postulate, which we now state and immediately check against Newton.

3.1 Hamilton's principle

Definition 3.1 (Lagrangian and action). For a mechanical system described by coordinates $\vec{q} = (q_1, \dots, q_n)$, the *Lagrangian* is

$$L(\vec{q}, \dot{\vec{q}}, t) = T - V \quad (38)$$

kinetic minus potential energy, and the *action* of a trajectory $\vec{q}(t)$ on $[t_0, t_1]$ is the functional

$$S[\vec{q}] = \int_{t_0}^{t_1} L(\vec{q}(t), \dot{\vec{q}}(t), t) dt. \quad (39)$$

Postulate 3.2 (Hamilton's principle of stationary action). The physical trajectory between fixed configurations $\vec{q}(t_0) = \vec{q}_A$ and $\vec{q}(t_1) = \vec{q}_B$ is a stationary point of the action: $\delta S = 0$.

By Theorem 2.1 in its several-function form (Remark 2.3, with $x \rightarrow t$), the postulate is equivalent to the *Euler–Lagrange equations of motion*

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \quad i = 1, \dots, n. \quad (40)$$

First duty: recover Newton. A postulate must first reproduce what is known. Take one particle in Cartesian coordinates, $V = V(\vec{r})$:

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - V(x, y, z). \quad (41)$$

For the x -equation:

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = m\ddot{x}, \quad \frac{\partial L}{\partial x} = -\frac{\partial V}{\partial x} = F_x, \quad (42)$$

so (40) reads $m\ddot{x} = F_x$, and likewise for y, z :

$$m\ddot{\vec{r}} = -\nabla V = \vec{F}. \quad (43)$$

Newton's second law, exactly. *For unconstrained conservative systems in Cartesian coordinates, Hamilton's principle and Newton's law are the same statement.* Everything the new formulation adds — and it adds a great deal — comes from the two theorems already in hand: covariance (Theorem 2.11: use any coordinates) and the treatment of constraints (next).

Remark 3.3 (Why $T - V$? The accounting, plus an intuition). The *minus* sign is not derivable from anything prior in this document: $L = T - V$ is part of the postulate, and its justification is exactly the check just performed — it is the choice that reproduces Newton. (Try $T + V$: the sign of the force flips.) That said, the combination can be *felt*. Take a thrown ball, and grant for intuition's sake that its short arcs *minimize* the action $\int (T - V) dt$ (true for sufficiently short arcs;

Convention 1.3 guards the general statement). The integrand is lowered by spending time where V is *high* — so the path bulges upward, lingering near the top of its flight — but height requires speed on the way up and down, and speed raises the kinetic term. Bulge too little and the $-V$ reduction is wasted; bulge too much and the kinetic cost dominates. The parabola is the balance point of this trade-off, and the Euler–Lagrange equation is the statement that the balance holds at every instant separately. This is intuition, not derivation — the derivation is the check above — but it is the right picture to carry.

3.2 Constraints and generalized coordinates

Real systems are full of constraints: the pendulum bob stays on its rod, the bead stays on its wire, the chain link stays attached to its neighbor. The Newtonian treatment must carry the *constraint forces* (rod tension, wire normal force) as unknowns and solve for them. The Lagrangian treatment makes them vanish from the outset. Here is the physics of why, kept in full in the main text because it *is* the key step.

Setup. Let a system of N particles $\vec{r}_a$ ($a = 1, \dots, N$) be subject to *holonomic* constraints — conditions on positions (possibly time), $f_\alpha(\vec{r}_1, \dots, \vec{r}_N, t) = 0$ — which are enforced by constraint forces $\vec{C}_a$, so that Newton reads

$$m_a \ddot{\vec{r}}_a = \vec{F}_a + \vec{C}_a \quad (\vec{F}_a = -\nabla_a V \text{ the applied forces}). \quad (44)$$

Suppose the constraint surface is smoothly parametrized by n *generalized coordinates*: $\vec{r}_a = \vec{r}_a(q_1, \dots, q_n, t)$.

Postulate 3.4 (Ideal constraints / d’Alembert). The constraint forces do no work under any *virtual displacement* — any infinitesimal displacement consistent with the constraints at frozen time:

$$\sum_{a=1}^N \vec{C}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = 0 \quad \text{for each } i. \quad (45)$$

This is a physical assumption, and an excellent one: rods, frictionless wires, and rigid links push only *perpendicular* to the motions they permit — that is what “rigid and frictionless” means. (Sliding friction violates it; such forces must be kept explicitly.)

The computation. Take Newton (44), dot each equation with $\partial \vec{r}_a / \partial q_i$, and sum over particles. The constraint forces drop by Postulate 3.4:

$$\sum_a m_a \ddot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \sum_a \vec{F}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} \equiv \mathcal{Q}_i \quad (\text{the generalized force}). \quad (46)$$

The left side is reorganized by two kinetic identities. From the chain rule, $\dot{\vec{r}}_a = \sum_j \frac{\partial \vec{r}_a}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_a}{\partial t}$, whence

$$(K1) \quad \frac{\partial \dot{\vec{r}}_a}{\partial \dot{q}_i} = \frac{\partial \vec{r}_a}{\partial q_i}, \quad (K2) \quad \frac{d}{dt} \frac{\partial \vec{r}_a}{\partial q_i} = \frac{\partial \dot{\vec{r}}_a}{\partial q_i} \quad (47)$$

— (K1) because $\dot{q}_i$ enters $\dot{\vec{r}}_a$ linearly with coefficient $\partial \vec{r}_a / \partial q_i$; (K2) because both sides equal $\sum_j \frac{\partial^2 \vec{r}_a}{\partial q_j \partial q_i} \dot{q}_j + \frac{\partial^2 \vec{r}_a}{\partial t \partial q_i}$ (mixed partials commute for C^2 maps). Now write, by the product rule,

$$\ddot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \frac{d}{dt} \left(\dot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} \right) - \dot{\vec{r}}_a \cdot \frac{d}{dt} \frac{\partial \vec{r}_a}{\partial q_i} \stackrel{(K1),(K2)}{=} \frac{d}{dt} \left(\dot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial \dot{q}_i} \right) - \dot{\vec{r}}_a \cdot \frac{\partial \dot{\vec{r}}_a}{\partial q_i}. \quad (48)$$

Multiply by m_a , sum over a , and recognize the kinetic energy $T = \sum_a \frac{1}{2} m_a \dot{\vec{r}}_a^2$ on both terms:

$$\sum_a m_a \ddot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial \dot{q}_i} = \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i}. \quad (49)$$

If moreover the applied forces are conservative, $\vec{F}_a = -\nabla_a V$ with $V = V(\vec{r}_1, \dots, \vec{r}_N, t)$, then by the chain rule $Q_i = -\partial V / \partial q_i$, and since V has no $\dot{q}$ -dependence, (46) assembles into

$$\frac{d}{dt} \frac{\partial(T - V)}{\partial \dot{q}_i} - \frac{\partial(T - V)}{\partial q_i} = 0. \quad (50)$$

Theorem 3.5 (Lagrange's equations for constrained systems). *For a system with ideal holonomic constraints and conservative applied forces, the generalized coordinates obey the Euler–Lagrange equations (40) of $L = T - V$, with the constraint forces nowhere appearing and the constraints automatically respected.*

The working recipe of Lagrangian mechanics, now fully justified:

Recipe. (1) Choose any n independent coordinates that label the allowed configurations. (2) Write T and V in them. (3) Form $L = T - V$ and turn the Euler–Lagrange crank. Constraint forces never enter; if you want them afterward (engineering loads!), reinstate the constrained coordinate with a multiplier — the multiplier *is* the force, the catenary's λ writ general.

3.3 Conservation laws

The variational structure converts symmetries into conserved quantities with almost no work. Three instances, in increasing generality.

Cyclic coordinates. Define the *generalized (canonical) momentum* conjugate to q_i :

$$p_i \equiv \frac{\partial L}{\partial \dot{q}_i} \quad (51)$$

If L does not contain q_i (q_i is *cyclic*), then (40) gives $\dot{p}_i = \partial L / \partial q_i = 0$:

$$q_i \text{ cyclic} \implies p_i \text{ conserved.} \quad (52)$$

(Proposition 2.5 in mechanical clothing. Example: a free particle, $L = \frac{1}{2} m \dot{x}^2$, has x cyclic and $p = m \dot{x}$ conserved.)

Energy. Define the *energy function*

$$h(\vec{q}, \dot{\vec{q}}, t) \equiv \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L. \quad (53)$$

Proposition 3.6. *Along any solution of (40), $\frac{dh}{dt} = -\frac{\partial L}{\partial t}$. In particular, if L has no explicit time dependence, h is conserved.*

Proof. Differentiate along the trajectory:

$$\begin{aligned} \frac{dh}{dt} &= \sum_i \ddot{q}_i \frac{\partial L}{\partial \dot{q}_i} + \sum_i \dot{q}_i \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \left[\sum_i \frac{\partial L}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \frac{\partial L}{\partial t} \right] \\ &= \sum_i \dot{q}_i \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} \right] - \frac{\partial L}{\partial t} = - \frac{\partial L}{\partial t}, \end{aligned} \quad (54)$$

the bracket vanishing by the equations of motion. (This is the Beltrami identity, Proposition 2.6, in n variables: “no explicit t ” plays the role of “no explicit x .”) Whether h equals $T + V$ is a separate question answered precisely in §6.3; and whether h “is” the Hamiltonian is a sharper question than it looks, answered precisely in Remark 6.2. $\square$

Noether’s theorem. The two facts above are shadows of one statement: *continuous symmetries of L produce conserved quantities.*

Theorem 3.7 (Noether, point-symmetry version). *Let $\vec{q} \mapsto \vec{Q}(\vec{q}, s)$ be a smooth one-parameter family of transformations with $\vec{Q}(\vec{q}, 0) = \vec{q}$, and suppose it leaves the Lagrangian invariant:*

$$L(\vec{Q}(\vec{q}, s), \frac{d}{dt} \vec{Q}(\vec{q}, s), t) = L(\vec{q}, \dot{\vec{q}}, t) \quad \text{for all } s. \quad (55)$$

Then, writing $\eta_i(\vec{q}) = \left. \frac{\partial Q_i}{\partial s} \right|_{s=0}$ for the infinitesimal generator, the quantity

$$I = \sum_i \frac{\partial L}{\partial \dot{q}_i} \eta_i(\vec{q}) \quad (56)$$

is conserved along every solution of the equations of motion.

Proof. Differentiate the invariance identity with respect to s at $s = 0$. Since along the transformed family $\left. \frac{\partial}{\partial s} \dot{Q}_i \right|_{s=0} = \dot{\eta}_i$ (the s - and t -derivatives commute for smooth families),

$$0 = \left. \frac{d}{ds} \right|_0 L = \sum_i \frac{\partial L}{\partial q_i} \eta_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \dot{\eta}_i. \quad (57)$$

So far this holds for any curve. Now evaluate *on a solution* and use the equations of motion to replace $\partial L / \partial q_i = \frac{d}{dt} (\partial L / \partial \dot{q}_i)$:

$$0 = \sum_i \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right] \eta_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \dot{\eta}_i = \frac{d}{dt} \left[\sum_i \frac{\partial L}{\partial \dot{q}_i} \eta_i \right] = \frac{dI}{dt}. \quad (58) \quad \square$$

Example 3.8 (The classic pairs). (i) *Translation:* $Q = q + s$ for a coordinate absent from L ; $\eta = 1$, $I = p$. Cyclic coordinates are Noether’s theorem for translations in configuration space. (ii) *Rotation:* a particle in a central potential, $L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - V(\sqrt{x^2 + y^2})$, is invariant under $(x, y) \mapsto (x \cos s - y \sin s, x \sin s + y \cos s)$; the generator is $\eta = (-y, x)$ and

$$I = m\dot{x}(-y) + m\dot{y}(x) = m(x\dot{y} - y\dot{x}) = L_z, \quad (59)$$

angular momentum. (iii) Time translation stands slightly apart — it shifts the curve’s parameter rather than its values, so it does not fit the hypothesis of Theorem 3.7 as stated — and its conserved quantity is exactly the h of Proposition 3.6, whose proof is the time-translation analogue of the argument above. The moral is uniform: **momentum is the charge of spatial translation, angular momentum of rotation, energy of time translation.**

Example 3.9 (Central forces: the effective one-dimensional problem). The cyclic-coordinate machinery at full power. A particle in a plane under $V = V(r)$ (planar without loss for a central force: angular momentum fixes the orbital plane [stated; it follows from conservation of the vector angular momentum $m \vec{r} \times \dot{\vec{r}}$]), in polar coordinates:

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) - V(r). \quad (60)$$

The angle is cyclic, so its momentum is conserved,

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = mr^2\dot{\varphi} \equiv \ell = \text{const}, \quad (61)$$

and the conserved energy $h = T + V$ (constraints time-independent), with $\dot{\varphi}$ eliminated via $\dot{\varphi} = \ell/mr^2$, becomes

$$E = \frac{1}{2}m\dot{r}^2 + \underbrace{\frac{\ell^2}{2mr^2}}_{V_{\text{eff}}(r)} + V(r) \quad (62)$$

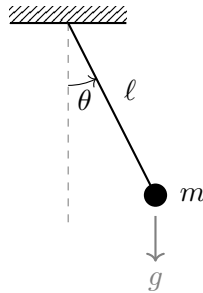
— a *one-dimensional* problem on the half-line $r > 0$, with the conserved angular momentum converted into a repulsive *centrifugal barrier* $\ell^2/2mr^2$. Circular orbits sit at $V'_{\text{eff}}(r_0) = 0$, stable where $V''_{\text{eff}} > 0$; the full radial motion is read off the one-dimensional energy diagram exactly as in §9. One warning, displayed because it is the step a reader gets wrong: eliminate $\dot{\varphi}$ in the *energy*, not in the Lagrangian — substituting $\dot{\varphi} = \ell/mr^2$ into L before varying flips the sign of the centrifugal term in the resulting equation of motion [the correct reduction at the Lagrangian level is the Routhian; stated]. (The quantum version of this reduction, with ℓ^2 replaced by $\ell(\ell + 1)\hbar^2$, is the radial equation of the quantum companion.)

4 The Pendulum

The recipe of §3.2, run start to finish on the simplest constrained system — one rigid rod, one angle. The point of the exercise is partly the result (the harmonic oscillator's most venerable realization) but mostly the *silence*: watch for what never appears in the calculation.

4.1 Setup and equation of motion

A bob of mass m on a rigid massless rod of length ℓ , pivoting freely in a vertical plane; uniform gravity g .



The constraint (fixed distance ℓ from the pivot) is holonomic and ideal (the rod's force lies along itself, perpendicular to the circular motion). One generalized coordinate suffices: the angle θ from the downward vertical. Cartesian positions and velocities:

$$x = \ell \sin \theta, \quad y = -\ell \cos \theta \quad \implies \quad \dot{x} = \ell \cos \theta \dot{\theta}, \quad \dot{y} = \ell \sin \theta \dot{\theta}, \quad (63)$$

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}m\ell^2\dot{\theta}^2(\cos^2 \theta + \sin^2 \theta) = \frac{1}{2}m\ell^2\dot{\theta}^2, \quad V = mgy = -mg\ell \cos \theta. \quad (64)$$

$$\boxed{L(\theta, \dot{\theta}) = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell \cos \theta} \quad (65)$$

Euler–Lagrange:

$$\frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta}, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = m\ell^2\ddot{\theta}, \quad \frac{\partial L}{\partial \theta} = -mg\ell \sin \theta, \quad (66)$$

$$\boxed{\ddot{\theta} = -\frac{g}{\ell} \sin \theta} \quad (67)$$

Note what did *not* happen: the rod tension was never introduced, never computed, never eliminated — the coordinate θ respects the constraint by construction, and Theorem 3.5 did the rest. (Compare the Newtonian route: decompose gravity along and across the rod, introduce the tension, project. Same result, more work, and the work multiplies with each additional constraint — as §5 is about to demonstrate by its absence.)

4.2 Small angles: the harmonic oscillator recovered

For $|\theta| \ll 1$, $\sin \theta = \theta + O(\theta^3)$, and (67) linearizes to

$$\ddot{\theta} = -\omega_0^2 \theta, \quad \omega_0 = \sqrt{\frac{g}{\ell}} \quad (68)$$

— the simple harmonic oscillator, whose complete theory (solution forms, energy, damping, driving, resonance) is the subject of the companion oscillator reference and is reused here wholesale. Period $T = 2\pi\sqrt{\ell/g}$, independent of amplitude and of mass.

Remark 4.1 (Beyond small angles). Equation (67) is exactly solvable, but not in elementary functions: conservation of h gives $\frac{1}{2}\ell\dot{\theta}^2 = g(\cos \theta - \cos \theta_{\max})$, and separating variables produces a period

$$T(\theta_{\max}) = 4\sqrt{\frac{\ell}{g}} K\left(\sin \frac{\theta_{\max}}{2}\right), \quad K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad (69)$$

a complete elliptic integral. Expanding: $T = 2\pi\sqrt{\ell/g} \left(1 + \frac{1}{16}\theta_{\max}^2 + O(\theta_{\max}^4)\right)$ — the period grows with amplitude, by about 1.7% at $\theta_{\max} = 30^\circ$. The isochronism of §4 is the θ^2 -truncation speaking, not the pendulum. (The curve that *is* exactly isochronous is the cycloid of §2.4 — the tautochrone; the subject's opening problem and its first mechanical example are secretly the same curve.)

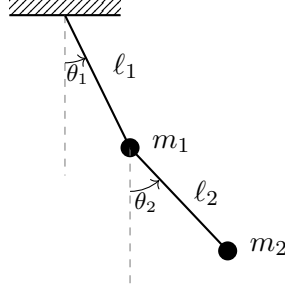
5 The Double Pendulum at Small Angles

The first system where the Lagrangian route is decisively easier. Attempt this system in Newtonian style and you will meet two unknown rod tensions, one of which acts on a bob that is itself accelerating in a curved path determined by the other — a genuinely unpleasant exercise in algebra. The Lagrangian route needs eleven lines of kinematics and two derivatives. The system is also the smallest one exhibiting two structural novelties: a kinetic energy that is *not* a sum of squares

(the mass matrix acquires off-diagonal terms), which will force the correct generalization of the eigenvalue problem; and, beyond small angles, deterministic chaos (§11) — all from two rods and two bobs.

5.1 Setup and kinematics

Bob 1 (mass m_1) hangs from the pivot on a rod of length ℓ_1 ; bob 2 (mass m_2) hangs from bob 1 on a rod of length ℓ_2 . Angles θ_1, θ_2 from the vertical.



Positions (y upward from the pivot):

$$x_1 = \ell_1 \sin \theta_1, \quad y_1 = -\ell_1 \cos \theta_1, \quad (70)$$

$$x_2 = \ell_1 \sin \theta_1 + \ell_2 \sin \theta_2, \quad y_2 = -\ell_1 \cos \theta_1 - \ell_2 \cos \theta_2. \quad (71)$$

Velocities by the chain rule:

$$\dot{x}_2 = \ell_1 \cos \theta_1 \dot{\theta}_1 + \ell_2 \cos \theta_2 \dot{\theta}_2, \quad \dot{y}_2 = \ell_1 \sin \theta_1 \dot{\theta}_1 + \ell_2 \sin \theta_2 \dot{\theta}_2, \quad (72)$$

and the speed of bob 2, expanded carefully (the cross term is the whole story of this system):

$$\begin{aligned} \dot{x}_2^2 + \dot{y}_2^2 &= \ell_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) + \ell_2^2 \dot{\theta}_2^2 (\cos^2 \theta_2 + \sin^2 \theta_2) \\ &\quad + 2\ell_1 \ell_2 \dot{\theta}_1 \dot{\theta}_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) \\ &= \ell_1^2 \dot{\theta}_1^2 + \ell_2^2 \dot{\theta}_2^2 + 2\ell_1 \ell_2 \cos(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2. \end{aligned} \quad (73)$$

Hence

$$T = \frac{1}{2} m_1 \ell_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 \left[\ell_1^2 \dot{\theta}_1^2 + \ell_2^2 \dot{\theta}_2^2 + 2\ell_1 \ell_2 \cos(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2 \right], \quad (74)$$

$$V = -(m_1 + m_2) g \ell_1 \cos \theta_1 - m_2 g \ell_2 \cos \theta_2, \quad (75)$$

and $L = T - V$. Pause on what was *not* needed: neither rod tension appears, no forces were decomposed, no free-body diagram of bob 2 in the accelerating frame of bob 1 was contemplated. The two constraints cost nothing.

5.2 Exact equations of motion

Write $\Delta \equiv \theta_1 - \theta_2$. For θ_1 :

$$\frac{\partial L}{\partial \dot{\theta}_1} = (m_1 + m_2) \ell_1^2 \dot{\theta}_1 + m_2 \ell_1 \ell_2 \cos \Delta \dot{\theta}_2, \quad (76)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} = (m_1 + m_2) \ell_1^2 \ddot{\theta}_1 + m_2 \ell_1 \ell_2 \cos \Delta \ddot{\theta}_2 - m_2 \ell_1 \ell_2 \sin \Delta (\dot{\theta}_1 - \dot{\theta}_2) \dot{\theta}_2, \quad (77)$$

$$\frac{\partial L}{\partial \theta_1} = -m_2 \ell_1 \ell_2 \sin \Delta \dot{\theta}_1 \dot{\theta}_2 - (m_1 + m_2) g \ell_1 \sin \theta_1, \quad (78)$$

and the Euler–Lagrange equation, after the two $\sin \Delta$ terms combine,

$$(m_1+m_2)\ell_1^2\ddot{\theta}_1 + m_2\ell_1\ell_2\cos\Delta\ddot{\theta}_2 + m_2\ell_1\ell_2\sin\Delta\dot{\theta}_2^2 + (m_1+m_2)g\ell_1\sin\theta_1 = 0. \quad (79)$$

For θ_2 , the same procedure ($\partial L/\partial\theta_2$ picks up $+m_2\ell_1\ell_2\sin\Delta\dot{\theta}_1\dot{\theta}_2$, note the sign) gives

$$m_2\ell_2^2\ddot{\theta}_2 + m_2\ell_1\ell_2\cos\Delta\ddot{\theta}_1 - m_2\ell_1\ell_2\sin\Delta\dot{\theta}_1^2 + m_2g\ell_2\sin\theta_2 = 0. \quad (80)$$

These exact equations are nonlinear, coupled, and — as §11 notes — chaotic at large amplitude. Our target is their small-angle limit.

5.3 Linearization done in the right order

Small oscillations about the hanging equilibrium: $\theta_i, \dot{\theta}_i$ all first-order small. There are two natural procedures — (a) linearize the exact equations (79)–(80), or (b) truncate L at quadratic order and *then* apply Euler–Lagrange — and it deserves one displayed sentence that they agree:

Lemma 5.1 (Expand-then-vary equals vary-then-expand). *The Euler–Lagrange equations of the quadratic truncation L_2 of L about an equilibrium coincide with the linearization of the Euler–Lagrange equations of L about that equilibrium.*

Proof. Euler–Lagrange takes one derivative of L in each slot; hence terms of cubic and higher order in $(\vec{\theta}, \dot{\vec{\theta}})$ in L produce terms of quadratic and higher order in the equations, which the linearization discards. Conversely every quadratic term of L survives one differentiation as a linear term. The two truncations remove exactly the same terms. $\square$

Route (b) is less error-prone (truncate once, differentiate a polynomial). Truncate: $\cos\Delta = 1 + O(\Delta^2)$ multiplies $\dot{\theta}_1\dot{\theta}_2$, already second order, so the correction is fourth order — drop it; $\cos\theta_i = 1 - \frac{1}{2}\theta_i^2 + O(\theta_i^4)$ in V ; discard constants. The quadratic Lagrangian:

$$L_2 = \frac{1}{2}(m_1+m_2)\ell_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\ell_2^2\dot{\theta}_2^2 + m_2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2 - \frac{1}{2}(m_1+m_2)g\ell_1\theta_1^2 - \frac{1}{2}m_2g\ell_2\theta_2^2. \quad (81)$$

Euler–Lagrange on this polynomial (each derivative now trivial):

$$\begin{aligned} (m_1+m_2)\ell_1^2\ddot{\theta}_1 + m_2\ell_1\ell_2\ddot{\theta}_2 &= -(m_1+m_2)g\ell_1\theta_1, \\ m_2\ell_1\ell_2\ddot{\theta}_1 + m_2\ell_2^2\ddot{\theta}_2 &= -m_2g\ell_2\theta_2, \end{aligned} \quad (82)$$

i.e., in matrix form,

$$\boxed{\mathbf{M}\ddot{\vec{\theta}} = -\mathbf{K}\vec{\theta}, \quad \mathbf{M} = \begin{pmatrix} (m_1+m_2)\ell_1^2 & m_2\ell_1\ell_2 \\ m_2\ell_1\ell_2 & m_2\ell_2^2 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} (m_1+m_2)g\ell_1 & 0 \\ 0 & m_2g\ell_2 \end{pmatrix}} \quad (83)$$

Both matrices are symmetric — $\mathbf{M}$ from the quadratic form T , $\mathbf{K}$ from the quadratic form V — and both are positive definite ($T > 0$ for any nonzero velocity; V has a strict minimum at hanging equilibrium). But unlike every system in the oscillator companion, *the mass matrix is not diagonal*: the kinetic cross term $\dot{\theta}_1\dot{\theta}_2$ couples the coordinates through their *velocities*. The plain eigenvalue problem no longer applies; its correct generalization is next.

5.4 The generalized eigenvalue problem

Insert the oscillatory ansatz $\vec{\theta}(t) = \vec{v} \cos(\omega t - \phi)$ into (83):

$$-\omega^2 \mathbf{M} \vec{v} \cos(\omega t - \phi) = -\mathbf{K} \vec{v} \cos(\omega t - \phi) \iff \boxed{\mathbf{K} \vec{v} = \omega^2 \mathbf{M} \vec{v}} \quad (84)$$

— a *generalized* eigenvalue problem: $\mathbf{M}$ stands where the identity used to. Nontrivial $\vec{v}$ requires

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0. \quad (85)$$

The general theory is exactly as strong as in the ordinary case:

Theorem 5.2 (Normal modes exist and diagonalize the system). *Let $\mathbf{M}$ be symmetric positive definite and $\mathbf{K}$ symmetric ($n \times n$, real). Then (84) has n real eigenvalues $\omega_1^2 \leq \dots \leq \omega_n^2$ (positive if $\mathbf{K}$ is positive definite) with eigenvectors $\vec{v}_1, \dots, \vec{v}_n$ that form a basis and are orthogonal in the $\mathbf{M}$ -inner product, $\vec{v}_r^\top \mathbf{M} \vec{v}_s = 0$ for $r \neq s$. In the coordinates $\vec{\theta} = \sum_r \xi_r \vec{v}_r$, the system (83) decouples into n independent harmonic oscillators $\ddot{\xi}_r = -\omega_r^2 \xi_r$.*

Proof in Appendix E; the idea is one line — since $\mathbf{M} > 0$ it has a symmetric square root, and the substitution $\vec{w} = \mathbf{M}^{1/2} \vec{v}$ turns (84) into an *ordinary* symmetric eigenvalue problem for $\mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2}$, whose spectral theorem does all the work. The physics of the statement: *small oscillations of any constrained conservative system about a stable equilibrium are superpositions of independent normal modes* — the oscillator companion’s central structure, now proved in the generality that mechanics actually delivers.

5.5 Equal masses and lengths: the explicit modes

Set $m_1 = m_2 = m$, $\ell_1 = \ell_2 = \ell$, and abbreviate $\lambda \equiv \omega^2 \ell / g$ (dimensionless). Then

$$\mathbf{M} = m\ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{K} = mg\ell \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{K} - \omega^2 \mathbf{M} = mg\ell \begin{pmatrix} 2 - 2\lambda & -\lambda \\ -\lambda & 1 - \lambda \end{pmatrix}. \quad (86)$$

Determinant:

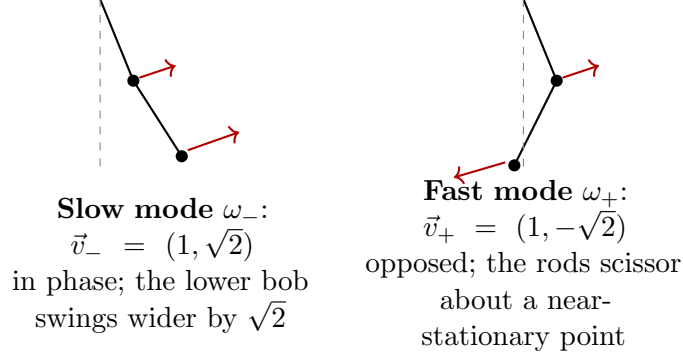
$$(2 - 2\lambda)(1 - \lambda) - \lambda^2 = 2 - 4\lambda + 2\lambda^2 - \lambda^2 = \lambda^2 - 4\lambda + 2 = 0 \implies \lambda_{\pm} = 2 \pm \sqrt{2}, \quad (87)$$

$$\boxed{\omega_-^2 = (2 - \sqrt{2}) \frac{g}{\ell} \approx 0.586 \frac{g}{\ell}, \quad \omega_+^2 = (2 + \sqrt{2}) \frac{g}{\ell} \approx 3.414 \frac{g}{\ell}} \quad (88)$$

Eigenvectors from the first row, $(2 - 2\lambda)v_1 = \lambda v_2$:

$$\frac{v_2}{v_1} = \frac{2 - 2\lambda}{\lambda} = \begin{cases} +\sqrt{2} & (\lambda = 2 - \sqrt{2}), \\ -\sqrt{2} & (\lambda = 2 + \sqrt{2}), \end{cases} \quad (89)$$

(for $\lambda_- = 2 - \sqrt{2}$: $\frac{2-2(2-\sqrt{2})}{2-\sqrt{2}} = \frac{2\sqrt{2}-2}{2-\sqrt{2}} = \frac{2(\sqrt{2}-1)(2+\sqrt{2})}{(2-\sqrt{2})(2+\sqrt{2})} = \frac{2(\sqrt{2}-1)(2+\sqrt{2})}{2} = (\sqrt{2}-1)(2+\sqrt{2}) = \sqrt{2}$; the other case is the mirror computation). Hence



Sanity checks, each one line: the frequencies bracket the single pendulum's, $\omega_-^2 < g/\ell < \omega_+^2$ (the in-phase mode is effectively a longer pendulum, the scissoring mode a stiffer one); M-orthogonality holds, $\vec{v}_-^T \mathbf{M} \vec{v}_+ = m\ell^2 (1, \sqrt{2}) \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -\sqrt{2} \end{pmatrix} = m\ell^2 (1, \sqrt{2}) \cdot (2 - \sqrt{2}, 1 - \sqrt{2})^T = m\ell^2 (2 - \sqrt{2} + \sqrt{2} - 2) = 0$ ✓ (note: plain orthogonality fails, $(1, \sqrt{2}) \cdot (1, -\sqrt{2}) = -1 \neq 0$ — the $\mathbf{M}$ -inner product is not decorative); and by Theorem 5.2, the general small motion is

$$\vec{\theta}(t) = \xi_-(0\text{-data}) \vec{v}_- \cos(\omega_- t - \phi_-) + \xi_+(0\text{-data}) \vec{v}_+ \cos(\omega_+ t - \phi_+), \quad (90)$$

with beats, energy exchange, and every phenomenon of the oscillator companion's coupled chapter available on demand.

Part II

Hamiltonian Mechanics

6 The Legendre Transform and Hamilton's Equations

Lagrangian mechanics runs on $(\vec{q}, \dot{\vec{q}})$: positions and velocities. Hamiltonian mechanics changes one variable — velocity out, momentum in — and before the machinery, the case for bothering. First, the *shape* of the dynamics changes: n second-order equations become $2n$ *first-order* ones, so that a single point in the $2n$ -dimensional space of $(\vec{q}, \vec{p})$ determines the entire future [existence and uniqueness for the smooth first-order system; stated] — the dynamics becomes a *flow*, and flows can be drawn, transported, and reasoned about globally (§9 will prove a theorem, Liouville's, that is invisible in the Lagrangian picture and is the opening axiom of statistical mechanics). Second, the choice of *momentum* as the fundamental variable is not arbitrary: p_i is what conservation laws conserve (§3.3), and in quantum mechanics it is $\hat{p}$, not $\hat{q}$, that stands beside $\hat{q}$ in the commutation relations — classical mechanics in Hamiltonian form is the version that quantizes. Third, and deepest: putting $\vec{q}$ and $\vec{p}$ on *equal footing* unlocks a group of transformations (mixing coordinates with momenta!) vastly larger than the point transformations of Theorem 2.11, and with it the strongest solution techniques classical mechanics possesses (§11). Goldstein's summary is the standard one: the Hamiltonian formulation's advantage is not calculational convenience but structural insight — it is the formal skeleton on which the more modern theories of matter are built. The change of variable itself is not a cosmetic substitution but a *Legendre transform*, whose structure is worth stating exactly.

6.1 The transform

Definition 6.1 (Legendre transform of the Lagrangian). Assume $L(\vec{q}, \dot{\vec{q}}, t)$ is C^2 and *convex in the velocities*: the Hessian $(\partial^2 L / \partial \dot{q}_i \partial \dot{q}_j)$ is positive definite. Define the *canonical momenta*

$$p_i \equiv \frac{\partial L}{\partial \dot{q}_i}(\vec{q}, \dot{\vec{q}}, t) \quad (91)$$

Convexity makes the map $\dot{\vec{q}} \mapsto \vec{p}$ invertible for each fixed $(\vec{q}, t)$ (its Jacobian is the positive-definite Hessian; the inverse function theorem applies), so velocities can be re-expressed as $\dot{\vec{q}} = \dot{\vec{q}}(\vec{q}, \vec{p}, t)$. The *Hamiltonian* is

$$H(\vec{q}, \vec{p}, t) \equiv \sum_i p_i \dot{q}_i(\vec{q}, \vec{p}, t) - L(\vec{q}, \dot{\vec{q}}(\vec{q}, \vec{p}, t), t). \quad (92)$$

For mechanical Lagrangians the convexity hypothesis is physical: the Hessian of T in the velocities is the mass matrix $M(\vec{q})$ of §5.4, positive definite because kinetic energy is positive.

6.2 Hamilton's equations by two routes

Derivation 1: track the differential. Compute dH from (92), treating H as a function of $(\vec{q}, \vec{p}, t)$ but differentiating through the definition:

$$\begin{aligned} dH &= \sum_i (p_i d\dot{q}_i + \dot{q}_i dp_i) - \sum_i \frac{\partial L}{\partial q_i} dq_i - \sum_i \underbrace{\frac{\partial L}{\partial \dot{q}_i}}_{=p_i} d\dot{q}_i - \frac{\partial L}{\partial t} dt \\ &= \sum_i \dot{q}_i dp_i - \sum_i \frac{\partial L}{\partial q_i} dq_i - \frac{\partial L}{\partial t} dt. \end{aligned} \quad (93)$$

The $d\dot{q}_i$ terms cancel *identically* — this cancellation is the entire design of the Legendre transform: it removes the dependence on the old variable and leaves a function of the new one. Reading off the coefficients of the independent differentials dq_i, dp_i, dt :

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \quad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \quad (94)$$

So far, no dynamics — these are identities of the transform. Dynamics enters through Euler–Lagrange: *on a solution*, $\partial L / \partial q_i = \frac{d}{dt}(\partial L / \partial \dot{q}_i) = \dot{p}_i$, and the middle identity becomes an equation of motion:

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}} \quad (i = 1, \dots, n) \quad (95)$$

— *Hamilton's equations*: $2n$ first-order equations replacing n second-order ones, in a form so symmetric that q and p differ only by a sign.

Derivation 2: a variational principle in phase space. Hamilton's equations are themselves Euler–Lagrange equations — of the *phase-space action*

$$S[\vec{q}, \vec{p}] = \int_{t_0}^{t_1} \left[\sum_i p_i \dot{q}_i - H(\vec{q}, \vec{p}, t) \right] dt, \quad (96)$$

in which $\vec{q}(t)$ and $\vec{p}(t)$ are varied *independently*, with the q 's pinned at the endpoints and the p 's entirely free. Vary $\vec{p}$ (no integration by parts needed, since $\dot{\vec{p}}$ never appears):

$$\delta_p S = \int \sum_i \left[\dot{q}_i - \frac{\partial H}{\partial p_i} \right] \delta p_i dt = 0 \quad \implies \quad \dot{q}_i = \frac{\partial H}{\partial p_i}, \quad (97)$$

by the fundamental lemma. Vary $\vec{q}$ (one integration by parts on $p_i \delta \dot{q}_i$; the boundary term dies on the pinned q 's):

$$\delta_q S = \int \sum_i \left[-\dot{p}_i - \frac{\partial H}{\partial q_i} \right] \delta q_i dt = 0 \quad \implies \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \quad (98)$$

Both equations from one functional, on equal footing — the formulation that generalizes cleanly (to field theory, and to the path integral).

Energy bookkeeping. Along a solution of (95),

$$\frac{dH}{dt} = \sum_i \left[\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right] + \frac{\partial H}{\partial t} = \sum_i \left[\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right] + \frac{\partial H}{\partial t} \implies \boxed{\frac{dH}{dt} = \frac{\partial H}{\partial t}} : (99)$$

if H has no explicit time dependence it is conserved.

Remark 6.2 (h versus H : the same number, not the same function). Comparing (92) with §3.3: along any state, H takes the *value* of the energy function h — $H(\vec{q}, \vec{p}, t) = h(\vec{q}, \vec{q}, t)$ whenever $\vec{p}$ and $\vec{q}$ correspond under (91). Are they then the same object? *No*, and the difference carries the whole formalism: they are different functions of *different variables* — h lives on $(\vec{q}, \vec{q})$, H on $(\vec{q}, \vec{p})$ — and partial derivatives care about what is held fixed. Concretely, differentiate $h = \sum_j \dot{q}_j \partial L / \partial \dot{q}_j - L$ at fixed $\vec{q}$ and compare with (94):

$$\left(\frac{\partial h}{\partial q_i} \right)_{\vec{q}} = \sum_j \dot{q}_j \frac{\partial^2 L}{\partial q_i \partial \dot{q}_j} - \frac{\partial L}{\partial q_i}, \quad \left(\frac{\partial H}{\partial q_i} \right)_{\vec{p}} = - \frac{\partial L}{\partial q_i} : (100)$$

the two disagree by $\sum_j \dot{q}_j \partial^2 L / \partial q_i \partial \dot{q}_j$ — zero for a constant mass matrix, but *nonzero* whenever M depends on the coordinates, as in the double pendulum. This is why Hamilton's equations hold for H and not for h : the clean identity $\partial H / \partial q_i = -\partial L / \partial q_i$ is a property of differentiating *at fixed momentum*, and arranging for it is exactly what the Legendre transform's cancellation in (93) accomplished. The same re-reading applies to the momentum itself: in §3.3, $p_i = \partial L / \partial \dot{q}_i$ was a *derived bookkeeping quantity* — a combination evaluated along a trajectory, singled out because cyclic coordinates conserve it. Here the identical formula is re-read as a *change of variables*, and p_i is promoted to an independent coordinate of state space. Same formula, different ontological status; the Legendre transform is the promotion ceremony, and keeping the two names (h on the Lagrangian side, H on the Hamiltonian side, following Goldstein) keeps the two objects distinct.

Remark 6.3 (Involutivity). The transform undoes itself: from (94), $\dot{q}_i = \partial H / \partial p_i$, and then $\sum_i p_i \dot{q}_i - H = L$ by (92) — performing the Legendre transform of H in the p 's returns L . Lagrangian and Hamiltonian mechanics are two charts on the same theory, with a lossless dictionary between them (given convexity).

6.3 When is $H = T + V$?

Proposition 6.4. *Suppose the constraints are time-independent (so that $\vec{r}_a = \vec{r}_a(\vec{q})$) and T is purely quadratic in the velocities, $T = \frac{1}{2} \sum_{ij} M_{ij}(\vec{q}) \dot{q}_i \dot{q}_j$ and $V = V(\vec{q}, t)$ is velocity-independent. Then $H = T + V$.*

Proof. The one tool needed is *Euler's theorem on homogeneous functions* in its quadratic case, proved inline: with $T = \frac{1}{2} \sum_{ij} M_{ij} \dot{q}_i \dot{q}_j$ and $M_{ij} = M_{ji}$,

$$\frac{\partial T}{\partial \dot{q}_k} = \sum_j M_{kj} \dot{q}_j \implies \sum_k \dot{q}_k \frac{\partial T}{\partial \dot{q}_k} = \sum_{kj} M_{kj} \dot{q}_k \dot{q}_j = 2T. (101)$$

Since V carries no velocities, $p_k = \partial T / \partial \dot{q}_k$, and

$$H = \sum_k p_k \dot{q}_k - L = 2T - (T - V) = T + V. (102)$$

□

Remark 6.5 (The rheonomic caveat). If the constraints move — $\vec{r}_a = \vec{r}_a(\vec{q}, t)$, e.g. a bead on a hoop rotating at imposed rate — then T acquires terms linear in $\dot{q}$ and a $\dot{q}$ -free term, the computation above breaks, and $H \neq T + V$. H may still be conserved (if L ends up t -independent) while the mechanical energy $T + V$ is *not* — whatever drives the constraint does work. The two statements “ H is conserved” and “ H is the energy” have different hypotheses; conflating them is the standard trap.

6.4 Examples

Harmonic oscillator. $L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2$; $p = m\dot{q}$, so $\dot{q} = p/m$ and

$$H = p \cdot \frac{p}{m} - \frac{p^2}{2m} + \frac{1}{2}kq^2 = \frac{p^2}{2m} + \frac{kq^2}{2}, \quad (103)$$

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial H}{\partial q} = -kq \implies \ddot{q} = \frac{\dot{p}}{m} = -\frac{k}{m}q = -\omega_0^2 q. \checkmark \quad (104)$$

The pendulum and the double pendulum deserve more than a paragraph: the next section re-derives and *completely solves* both in the Hamiltonian formalism, as the proper stress test of the new machinery.

7 The Two Systems Revisited: Complete Hamiltonian Solutions

Part I solved the pendulum and the double pendulum by the Lagrangian recipe. We now solve both again, start to finish, in the Hamiltonian formalism — not as repetition but as calibration: the same physics must come out, and watching *where the same content sits in the two formalisms* is how a formalism is actually learned.

7.1 Complete solution of the pendulum

Setting up. From §4, $L = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell \cos \theta$. The canonical momentum and its inversion:

$$p = \frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta} \iff \dot{\theta} = \frac{p}{m\ell^2} \quad (105)$$

— physically, p is the *angular momentum* about the pivot (moment of inertia $m\ell^2$ times angular velocity); the canonical momentum conjugate to an angle is always an angular momentum. The Hamiltonian, via (92) or directly via Proposition 6.4 ($T + V$; the constraint is time-independent):

$$\boxed{H(\theta, p) = \frac{p^2}{2m\ell^2} - mg\ell \cos \theta} \quad (106)$$

Hamilton’s equations (95):

$$\dot{\theta} = \frac{\partial H}{\partial p} = \frac{p}{m\ell^2}, \quad \dot{p} = -\frac{\partial H}{\partial \theta} = -mg\ell \sin \theta. \quad (107)$$

(Consistency: differentiate the first equation and substitute the second to recover $\ddot{\theta} = -(g/\ell) \sin \theta$, equation (67). The same physics redistributed: one second-order equation has become a pair of first-order ones, and the pair is what draws the phase portrait of §9.)

Small angles: the complete solution. Expand H to quadratic order ($\cos \theta = 1 - \frac{1}{2}\theta^2 + O(\theta^4)$; drop the constant):

$$H_2 = \frac{p^2}{2m\ell^2} + \frac{mg\ell}{2}\theta^2, \quad \dot{\theta} = \frac{p}{m\ell^2}, \quad \dot{p} = -mg\ell\theta. \quad (108)$$

Differentiating the first equation and inserting the second gives $\ddot{\theta} = -(g/\ell)\theta = -\omega_0^2\theta$; then p follows from $p = m\ell^2\dot{\theta}$. With initial data (θ_0, p_0) :

$$\boxed{\theta(t) = \theta_0 \cos \omega_0 t + \frac{p_0}{m\ell^2 \omega_0} \sin \omega_0 t, \quad p(t) = p_0 \cos \omega_0 t - m\ell^2 \omega_0 \theta_0 \sin \omega_0 t} \quad (109)$$

(Checks: at $t = 0$ the pair returns (θ_0, p_0) ✓; substituting both expressions into (107) verifies the equations at all times, term by term ✓.) *Every* small motion of the pendulum is (109) — completeness leans on uniqueness for the linear system, which the energy method proves in three lines: the difference of two solutions with the same initial data starts with $H_2 = 0$, keeps H_2 conserved, and H_2 is positive definite, so the difference vanishes identically. The problem is closed.

Geometry of the solution: eliminate t . Direct substitution (using $\cos^2 + \sin^2 = 1$) shows $\frac{1}{2m\ell^2}p(t)^2 + \frac{mg\ell}{2}\theta(t)^2 = H_2(\theta_0, p_0) \equiv E$ for all t , so the orbit is the ellipse

$$\frac{\theta^2}{2E/mg\ell} + \frac{p^2}{2m\ell^2 E} = 1 \quad (110)$$

with semi-axes $\sqrt{2E/mg\ell}$ and $\sqrt{2m\ell^2 E}$, traversed once per period — precisely the small ovals around the centers of Fig. 3. Its enclosed area is π times the product of the semi-axes,

$$\text{area} = \pi \sqrt{\frac{2E}{mg\ell}} \sqrt{2m\ell^2 E} = 2\pi E \sqrt{\frac{\ell}{g}} = \frac{2\pi E}{\omega_0} \quad (111)$$

— area = $2\pi \times$ energy over frequency. The quantity E/ω_0 (enclosed area over 2π) is the *action variable* of Hamilton–Jacobi theory (§10, Remark 10.15), making its first, unforced appearance.

Large angles: complete reduction to quadrature. Beyond small angles the Hamiltonian route still delivers everything short of elementary functions, in three moves. (i) *Conservation solves for p* : on the energy surface $H = E$,

$$p(\theta) = \pm \sqrt{2m\ell^2 (E + mg\ell \cos \theta)}. \quad (112)$$

(ii) *The first Hamilton equation separates*: from $\dot{\theta} = p/m\ell^2$,

$$t(\theta) = \int \frac{m\ell^2 d\theta}{\sqrt{2m\ell^2 (E + mg\ell \cos \theta)}} \quad (113)$$

— the dynamics is *reduced to quadrature*: one integral stands between the initial data and the full trajectory. (iii) *The period, explicitly*. For libration with turning angle $\theta_{\max}$ (where $p = 0$): $E = -mg\ell \cos \theta_{\max}$, and a quarter period runs from $\theta = 0$ to $\theta_{\max}$:

$$\frac{T}{4} = \sqrt{\frac{\ell}{2g}} \int_0^{\theta_{\max}} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_{\max}}}. \quad (114)$$

Convert with $\cos \theta = 1 - 2 \sin^2(\theta/2)$, so that $\cos \theta - \cos \theta_{\max} = 2(\sin^2 \frac{\theta_{\max}}{2} - \sin^2 \frac{\theta}{2})$; then substitute $\sin \frac{\theta}{2} = k \sin \varphi$ with $k \equiv \sin \frac{\theta_{\max}}{2}$ (so $\varphi : 0 \rightarrow \pi/2$ as $\theta : 0 \rightarrow \theta_{\max}$), whence $\frac{1}{2} \cos \frac{\theta}{2} d\theta = k \cos \varphi d\varphi$ and $\cos \frac{\theta}{2} = \sqrt{1 - k^2 \sin^2 \varphi}$:

$$\frac{T}{4} = \sqrt{\frac{\ell}{2g}} \int_0^{\pi/2} \frac{1}{\sqrt{2} \sqrt{1 - k^2 \sin^2 \varphi}} \cdot \frac{2k \cos \varphi d\varphi}{k \cos \varphi} \implies \boxed{T = 4 \sqrt{\frac{\ell}{g}} K\left(\sin \frac{\theta_{\max}}{2}\right)} \quad (115)$$

— the elliptic-integral period stated in Remark 4.1, now derived. Every qualitative feature of the phase portrait becomes quantitative here: $K(k) \rightarrow \pi/2$ as $k \rightarrow 0$ recovers $T = 2\pi\sqrt{\ell/g}$, and $K(k) \rightarrow \infty$ as $k \rightarrow 1$ ($\theta_{\max} \rightarrow \pi$) is the diverging period at the separatrix.

7.2 Complete solution of the double pendulum

The Hamiltonian for any quadratic kinetic energy. One structural proposition organizes everything.

Proposition 7.1. *If $L = \frac{1}{2} \dot{\vec{q}}^T \mathbf{M}(\vec{q}) \dot{\vec{q}} - V(\vec{q})$ with $\mathbf{M}(\vec{q})$ symmetric positive definite, then*

$$\vec{p} = \mathbf{M}(\vec{q}) \dot{\vec{q}}, \quad \dot{\vec{q}} = \mathbf{M}(\vec{q})^{-1} \vec{p}, \quad \boxed{H(\vec{q}, \vec{p}) = \frac{1}{2} \vec{p}^T \mathbf{M}(\vec{q})^{-1} \vec{p} + V(\vec{q})}. \quad (116)$$

Proof. $p_i = \partial L / \partial \dot{q}_i = \sum_j M_{ij} \dot{q}_j$ is $\vec{p} = \mathbf{M} \dot{\vec{q}}$; positive definiteness makes $\mathbf{M}$ invertible, giving the middle display. Then, using the symmetry of $\mathbf{M}^{-1}$,

$$H = \vec{p}^T \dot{\vec{q}} - L = \vec{p}^T \mathbf{M}^{-1} \vec{p} - \frac{1}{2} (\mathbf{M}^{-1} \vec{p})^T \mathbf{M} (\mathbf{M}^{-1} \vec{p}) + V = \frac{1}{2} \vec{p}^T \mathbf{M}^{-1} \vec{p} + V. \quad (117)$$

(This is Proposition 6.4 again — $H = T + V$ — with T now written in the momenta.) $\square$

The exact double-pendulum Hamiltonian. From §5, with $\Delta = \theta_1 - \theta_2$,

$$\mathbf{M}(\vec{\theta}) = \begin{pmatrix} (m_1+m_2)\ell_1^2 & m_2\ell_1\ell_2 \cos \Delta \\ m_2\ell_1\ell_2 \cos \Delta & m_2\ell_2^2 \end{pmatrix}, \quad (118)$$

whose determinant deserves one display of its own:

$$\det \mathbf{M} = (m_1+m_2) m_2 \ell_1^2 \ell_2^2 - m_2^2 \ell_1^2 \ell_2^2 \cos^2 \Delta = m_2 \ell_1^2 \ell_2^2 (m_1 + m_2 \sin^2 \Delta) > 0 \quad (119)$$

for *all* configurations — the mass matrix never degenerates (even with the rods aligned, $\sin \Delta = 0$, the m_1 term keeps it positive), so the Legendre transform is globally legitimate. Inverting the 2×2 matrix and applying Proposition 7.1:

$$\boxed{H = \frac{m_2 \ell_2^2 p_1^2 + (m_1+m_2) \ell_1^2 p_2^2 - 2m_2 \ell_1 \ell_2 \cos \Delta p_1 p_2}{2 m_2 \ell_1^2 \ell_2^2 (m_1 + m_2 \sin^2 \Delta)} - (m_1+m_2)g\ell_1 \cos \theta_1 - m_2 g \ell_2 \cos \theta_2} \quad (120)$$

with $p_1 = (m_1+m_2)\ell_1^2 \dot{\theta}_1 + m_2 \ell_1 \ell_2 \cos \Delta \dot{\theta}_2$ and $p_2 = m_2 \ell_2^2 \dot{\theta}_2 + m_2 \ell_1 \ell_2 \cos \Delta \dot{\theta}_1$. The exact dynamics is the four first-order equations

$$\dot{\vec{\theta}} = \mathbf{M}(\vec{\theta})^{-1} \vec{p}, \quad \dot{\vec{p}} = - \frac{\partial}{\partial \vec{\theta}} \left[\frac{1}{2} \vec{p}^T \mathbf{M}(\vec{\theta})^{-1} \vec{p} \right] - \frac{\partial V}{\partial \vec{\theta}}, \quad (121)$$

where the configuration-dependence of the inverse is differentiated via $\partial_{\theta} \mathbf{M}^{-1} = -\mathbf{M}^{-1}(\partial_{\theta} \mathbf{M}) \mathbf{M}^{-1}$ (differentiate $\mathbf{M} \mathbf{M}^{-1} = \mathbb{I}$ by the product rule and solve). These four equations are the complete dynamical content of the system — what a numerical integrator integrates, and the flow whose chaos §11 describes.

Small angles: decoupling and solving Hamilton's equations. Quadratic truncation: in the kinetic term $\vec{p}$ is already first-order small, so $M(\vec{\theta})^{-1}$ may be evaluated at equilibrium ($\cos \Delta \rightarrow 1$; the correction enters at fourth order), giving the constant matrix M_0 of §5.3; the potential truncates as before. With K from (83),

$$H_2 = \frac{1}{2} \vec{p}^T M_0^{-1} \vec{p} + \frac{1}{2} \vec{\theta}^T K \vec{\theta}, \quad \dot{\vec{\theta}} = M_0^{-1} \vec{p}, \quad \dot{\vec{p}} = -K \vec{\theta}. \quad (122)$$

Now the payoff of Theorem 5.2. Take the generalized eigenvectors $K \vec{v}_r = \omega_r^2 M_0 \vec{v}_r$, normalized to $\vec{v}_r^T M_0 \vec{v}_s = \delta_{rs}$, and define *mode coordinates and mode momenta*

$$\xi_r \equiv \vec{v}_r^T M_0 \vec{\theta}, \quad \pi_r \equiv \vec{v}_r^T \vec{p}. \quad (123)$$

Differentiate each along the flow (122):

$$\dot{\xi}_r = \vec{v}_r^T M_0 \dot{\vec{\theta}} = \vec{v}_r^T M_0 M_0^{-1} \vec{p} = \vec{v}_r^T \vec{p} = \pi_r, \quad (124)$$

$$\dot{\pi}_r = \vec{v}_r^T \dot{\vec{p}} = -\vec{v}_r^T K \vec{\theta} = -\sum_s \xi_s \vec{v}_r^T K \vec{v}_s = -\omega_r^2 \xi_r \quad (125)$$

(the last step expands $\vec{\theta} = \sum_s \xi_s \vec{v}_s$ — valid because the $\vec{v}_s$ are a basis with coefficients recovered by M_0 -orthonormality — and uses $\vec{v}_r^T K \vec{v}_s = \omega_s^2 \vec{v}_r^T M_0 \vec{v}_s = \omega_s^2 \delta_{rs}$). Each pair obeys

$$\dot{\xi}_r = \pi_r, \quad \dot{\pi}_r = -\omega_r^2 \xi_r \quad (126)$$

— Hamilton's equations of the unit-mass oscillator Hamiltonian $\frac{1}{2} \pi_r^2 + \frac{1}{2} \omega_r^2 \xi_r^2$; and indeed substituting the expansions into H_2 collapses it (same two identities) to

$$H_2 = \sum_r \left[\frac{1}{2} \pi_r^2 + \frac{1}{2} \omega_r^2 \xi_r^2 \right] \quad (127)$$

— *two independent oscillators, additively*. Where the Lagrangian treatment decoupled the *equations* (Theorem 5.2), the Hamiltonian treatment decouples the *energy itself*: the modes carry separately conserved energies ($\dot{H}_r = 0$ each, by the one-oscillator computation). That the change of variables $(\vec{\theta}, \vec{p}) \rightarrow (\xi, \pi)$ preserves the Hamiltonian form of the equations makes it our first nontrivial *canonical transformation*; §8 will supply the certificate that verifies this in one line.

The complete general solution. Each mode is solved by (109) with $m\ell^2 \rightarrow 1$, $\omega_0 \rightarrow \omega_r$:

$$\xi_r(t) = \xi_r(0) \cos \omega_r t + \frac{\pi_r(0)}{\omega_r} \sin \omega_r t, \quad \pi_r(t) = \pi_r(0) \cos \omega_r t - \omega_r \xi_r(0) \sin \omega_r t, \quad (128)$$

and translating back through the definitions (using $\vec{p}(0) = M_0 \dot{\vec{\theta}}(0)$, so $\pi_r(0) = \vec{v}_r^T M_0 \dot{\vec{\theta}}(0)$),

$$\vec{\theta}(t) = \sum_r \left[\left(\vec{v}_r^T M_0 \vec{\theta}(0) \right) \cos \omega_r t + \frac{\vec{v}_r^T M_0 \dot{\vec{\theta}}(0)}{\omega_r} \sin \omega_r t \right] \vec{v}_r \quad (129)$$

Four numbers in, one motion out. For equal masses and lengths everything is explicit: $\omega_{\mp}^2 = (2 \mp \sqrt{2}) g/\ell$, unnormalized modes $(1, \pm\sqrt{2})$, with M_0 -norms

$$(1, \sqrt{2}) M_0 \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} = m\ell^2(4 + 2\sqrt{2}), \quad (1, -\sqrt{2}) M_0 \begin{pmatrix} 1 \\ -\sqrt{2} \end{pmatrix} = m\ell^2(4 - 2\sqrt{2}), \quad (130)$$

so $\vec{v}_\mp = (1, \pm\sqrt{2})/\sqrt{m\ell^2(4 \pm 2\sqrt{2})}$ in (129). One concrete instance, fully evaluated: release from rest with the upper rod displaced, $\vec{\theta}(0) = (\alpha, 0)$, $\dot{\vec{\theta}}(0) = 0$. The mode amplitudes are $\vec{v}_\mp^\top \mathbf{M}_0 \vec{\theta}(0)$ with $\mathbf{M}_0 \vec{\theta}(0) = \alpha m\ell^2 (2, 1)^\top$, and assembling (129) (all $m\ell^2$ factors cancel between the amplitude and the normalized eigenvector):

$$\theta_1(t) = \frac{\alpha}{2} [\cos \omega_- t + \cos \omega_+ t], \quad \theta_2(t) = \frac{\alpha}{\sqrt{2}} [\cos \omega_- t - \cos \omega_+ t] \quad (131)$$

(equivalently: expanding the initial condition in the eigenbasis, $(\alpha, 0) = \frac{\alpha}{2}(1, \sqrt{2}) + \frac{\alpha}{2}(1, -\sqrt{2})$, each mode released from rest evolves with its own cosine). Checks: $\theta_1(0) = \alpha$ ✓; $\theta_2(0) = 0$ ✓; both initial velocities vanish since every term is a cosine ✓; and the initial *accelerations* — the check sensitive to the mode weights — match Newton: differentiating twice at $t = 0$ gives $\ddot{\vec{\theta}}(0) = (-\frac{\alpha}{2}(\omega_-^2 + \omega_+^2), -\frac{\alpha}{\sqrt{2}}(\omega_-^2 - \omega_+^2)) = (-2\alpha g/\ell, +2\alpha g/\ell)$, in agreement with $\ddot{\vec{\theta}}(0) = -\mathbf{M}_0^{-1} \mathbf{K} \vec{\theta}(0)$ evaluated directly ✓. The lower bob (131) is a difference of cosines at frequency ratio $\omega_+/\omega_- = \sqrt{(2+\sqrt{2})/(2-\sqrt{2})} = 1 + \sqrt{2}$ — irrational: by the product identity of the oscillator companion the motion beats, energy migrating between the bobs, and because the ratio is irrational the motion never exactly repeats. Complete, explicit, checkable: the Hamiltonian route closes on the same physics the Lagrangian one opened.

8 Poisson Brackets

Hamilton's equations treat $\vec{q}$ and $\vec{p}$ symmetrically; the Poisson bracket is the algebraic structure that makes the symmetry total. The motivation is threefold: it packages the dynamics of *every* observable in one formula (not just coordinates and momenta); it converts questions about conservation into questions of pure algebra; and it supplies the practical certificate for *canonical* changes of variables — the one promised in §7.2. It is also, nearly symbol for symbol, the classical face of the quantum commutator.

8.1 Definition and algebra

Definition 8.1. For C^1 functions f, g on phase space (possibly time-dependent),

$$\{f, g\} \equiv \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right). \quad (132)$$

Three properties are immediate from the definition, one is not:

1. *Bilinearity*: $\{af + bg, h\} = a\{f, h\} + b\{g, h\}$ (a, b constants) — derivatives are linear.
2. *Antisymmetry*: $\{f, g\} = -\{g, f\}$; in particular $\{f, f\} = 0$ — swap the two terms.
3. *Leibniz rule*: $\{fg, h\} = f\{g, h\} + \{f, h\}g$ — apply the product rule inside each partial derivative of fg and regroup: $\partial_i(fg) = f\partial_i g + (\partial_i f)g$ in both slots. The bracket with h acts as a *derivation*.
4. *Jacobi identity*:

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0. \quad (133)$$

Not obvious: each summand contains second derivatives, and the identity asserts a threefold cancellation. The full computation is Appendix F; the organizing idea (worth knowing) is to sort all terms by *which* function carries the second derivative and check that each of the three groups cancels internally, using only the symmetry of mixed partials.

The canonical variables themselves obey the *fundamental brackets* (direct evaluation: $\partial q_i / \partial q_j = \delta_{ij}$ etc.):

$$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \boxed{\{q_i, p_j\} = \delta_{ij}} \quad (134)$$

Example 8.2 (The certificate promised in §7.2). A change of variables is *canonical* — preserves the Hamiltonian form of the equations for every H — precisely when the new variables reproduce the fundamental brackets (we use the criterion here in advance; §10.1 states and proves it in the direction this example needs). For the mode variables of the double pendulum, $\xi_r = \vec{v}_r^\top \mathbf{M}_0 \vec{\theta}$, $\pi_r = \vec{v}_r^\top \vec{p}$:

$$\{\xi_r, \pi_s\} = \sum_i \left[\frac{\partial \xi_r}{\partial \theta_i} \frac{\partial \pi_s}{\partial p_i} - \underbrace{\frac{\partial \xi_r}{\partial p_i} \frac{\partial \pi_s}{\partial \theta_i}}_{=0} \right] = \sum_i (\mathbf{M}_0 \vec{v}_r)_i (\vec{v}_s)_i = \vec{v}_s^\top \mathbf{M}_0 \vec{v}_r = \delta_{rs}, \quad (135)$$

and $\{\xi_r, \xi_s\} = \{\pi_r, \pi_s\} = 0$ similarly (each variable of a pair depends on only one half of the phase-space coordinates). The $\mathbf{M}_0$ -orthonormality of the modes — which entered §5.4 as linear algebra — is revealed as exactly the condition that mode variables are canonical.

8.2 Dynamics as bracket flow

Proposition 8.3. *Along any solution of Hamilton's equations, every C^1 phase-space function $f(\vec{q}, \vec{p}, t)$ evolves by*

$$\boxed{\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}}. \quad (136)$$

Proof. Chain rule along the trajectory, then Hamilton's equations (95):

$$\frac{df}{dt} = \sum_i \left[\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right] + \frac{\partial f}{\partial t} = \sum_i \left[\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right] + \frac{\partial f}{\partial t} = \{f, H\} + \frac{\partial f}{\partial t}. \quad \square \quad (137)$$

Special cases assemble the whole formalism into one notation: taking $f = q_i$ and $f = p_i$ recovers Hamilton's equations themselves,

$$\dot{q}_i = \{q_i, H\}, \quad \dot{p}_i = \{p_i, H\} \quad (138)$$

— coordinates and momenta on *identical* footing; taking $f = H$ recovers $dH/dt = \partial H / \partial t$ (antisymmetry kills $\{H, H\}$); and conservation acquires its final form:

$$f \text{ (time-independent) is conserved} \iff \{f, H\} = 0. \quad (139)$$

Corollary 8.4 (Poisson's theorem). *If f and g are conserved (time-independent, $\{f, H\} = \{g, H\} = 0$), so is $\{f, g\}$.*

Proof. By Jacobi with $h = H$: $\{\{f, g\}, H\} = -\{\{g, H\}, f\} - \{\{H, f\}, g\} = -\{0, f\} - \{0, g\} = 0$ (after shuffling with antisymmetry). New conserved quantities from old, by pure algebra — occasionally the bracket closes on the old ones (angular momenta: $\{L_x, L_y\} = L_z$), occasionally it produces genuinely new information. $\square$

8.3 Noether revisited: conserved quantities generate their symmetries

Noether's theorem (§3.3) was a one-way street: symmetry in, conserved quantity out. The Hamiltonian formalism reveals the street runs both ways — and that the two directions are the *same equation*.

The key construction: any time-independent phase-space function $G(\vec{q}, \vec{p})$ can be installed in place of H and used to generate a *flow* — a one-parameter family of transformations obtained by solving Hamilton's equations with G as the generator,

$$\frac{dq_i}{ds} = \frac{\partial G}{\partial p_i}, \quad \frac{dp_i}{ds} = -\frac{\partial G}{\partial q_i} \quad \left(\text{so that } \frac{df}{ds} = \{f, G\} \text{ for any } f \right). \quad (140)$$

Time evolution is then just the special case $G = H$, $s = t$. Now read one bracket in two directions:

$$\boxed{\begin{array}{ccc} \underbrace{\{G, H\} = 0} & \Longleftrightarrow & \underbrace{\{H, G\} = 0} \\ G \text{ conserved under time evolution} & & H \text{ invariant under the flow of } G \end{array}} \quad (141)$$

— the equivalence is the antisymmetry of the bracket, nothing more. *A quantity is conserved precisely when the transformation it generates is a symmetry of the Hamiltonian.* Symmetry and conservation, which Noether's theorem related by a computation, are exposed as one relation viewed from its two ends.

The circle closes on §3.3 exactly: given a point symmetry with generator $\eta_i(\vec{q})$, Noether's conserved quantity was $I = \sum_i p_i \eta_i(\vec{q})$ — and the flow that *this* I generates is

$$\frac{dq_i}{ds} = \frac{\partial I}{\partial p_i} = \eta_i(\vec{q}), \quad (142)$$

the original symmetry transformation itself. Check two instances:

- $G = p$ (one degree of freedom): flow $dq/ds = 1$, $dp/ds = 0$, i.e. $q \mapsto q + s$ — *momentum generates translations*.
- $G = L_z = xp_y - yp_x$: flow $dx/ds = \partial L_z / \partial p_x = -y$, $dy/ds = \partial L_z / \partial p_y = x$ — the infinitesimal rotation, with generator $(-y, x)$ matching the η of the Noether example in §3.3 — *angular momentum generates rotations*.

This is why conservation laws are placed where they are in this document: the theorem and its proof live on the Lagrangian side, where symmetries of L are stated; but their deepest formulation — observables as generators, symmetry $\Leftrightarrow$ conservation as a single antisymmetric bracket — is Hamiltonian property, and it is the form that survives into quantum mechanics ($\hat{p}$ generates translations, $\hat{L}_z$ generates rotations, $\hat{H}$ generates time evolution).

Remark 8.5 (The structure that survives quantization). The triple (bilinear, antisymmetric, Jacobi) makes phase-space functions a *Lie algebra*, with the Leibniz rule tying the algebra to the geometry. Quantum mechanics keeps this entire structure and changes one thing: $\{f, g\} \rightarrow \frac{1}{i\hbar}[\hat{f}, \hat{g}]$. In particular $\{q, p\} = 1$ becomes $[\hat{q}, \hat{p}] = i\hbar$. The bracket is not a notational convenience; it is the load-bearing wall.

9 Phase Space

9.1 Portraits

The state of the system is the point $(\vec{q}, \vec{p})$; Hamilton's equations define a *flow* on this $2n$ -dimensional space, and for one degree of freedom the entire global behavior can be drawn: since H is conserved, *trajectories are level curves of H* , and drawing dynamics reduces to drawing contour lines.

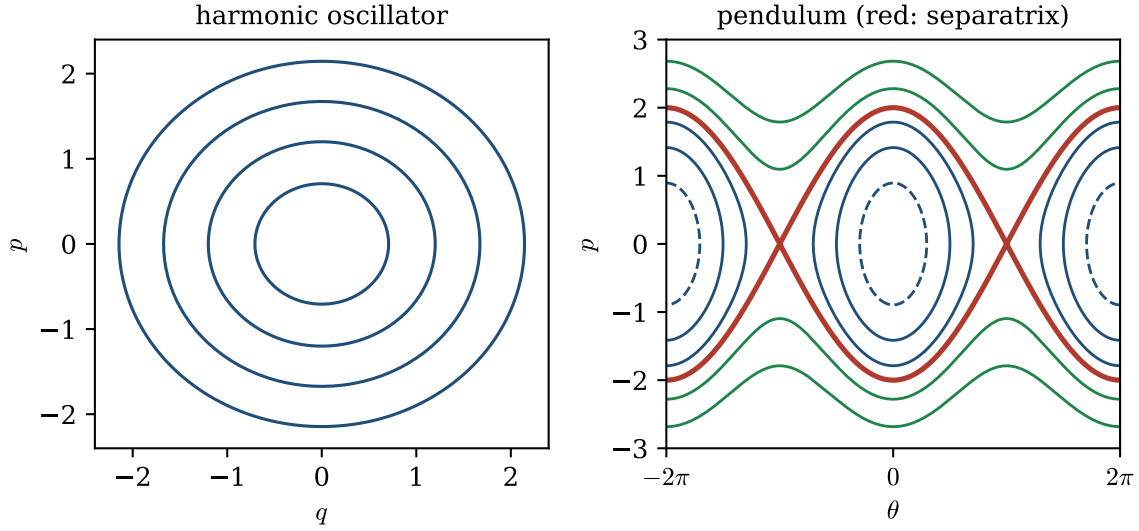


Figure 3: Left: the oscillator, $H = p^2/2m + kq^2/2$ — nested ellipses, one period each. Right: the pendulum (106) in normalized variables — closed curves (libration) inside the red *separatrix*, wavy open curves (rotation) outside it.

Read the pendulum portrait as a summary of everything the system can do:

- **Centers:** $(\theta, p) = (0, 0)$ (and its 2π translates) are stable equilibria; nearby level curves are small ellipses — the small-angle oscillator of §4, literally visible as the elliptical shape of low-energy contours.
- **Saddle:** $(\pm\pi, 0)$, the inverted pendulum, is an unstable equilibrium; level curves near it form an X.
- **Separatrix** (red): the level set $H = +mgl$ through the saddles — the watershed between *libration* (back-and-forth, closed curves) and *rotation* (over the top, open curves). A motion *on* the separatrix approaches the inverted position forever without arriving; the period of libration diverges as the amplitude approaches π (the elliptic K of Remark 4.1 blowing up as $k \rightarrow 1$).

One conserved function, one picture, the full qualitative theory — the economy that makes the Hamiltonian viewpoint the natural language for global and long-time questions.

9.2 Liouville's theorem

The flow does not merely move points; it moves them *incompressibly*.

Theorem 9.1 (Liouville). *The Hamiltonian phase flow preserves phase-space volume: if a region Ω_0 of initial conditions is carried by the flow to Ω_t , then $\text{vol}(\Omega_t) = \text{vol}(\Omega_0)$.*

Proof. The flow is generated by the phase-space velocity field

$$X = (\dot{q}_1, \dots, \dot{q}_n, \dot{p}_1, \dots, \dot{p}_n) = \left(\frac{\partial H}{\partial p_1}, \dots, \frac{\partial H}{\partial p_n}, -\frac{\partial H}{\partial q_1}, \dots, -\frac{\partial H}{\partial q_n} \right). \quad (143)$$

Its divergence vanishes *identically*:

$$\nabla \cdot X = \sum_{i=1}^n \left[\frac{\partial}{\partial q_i} \frac{\partial H}{\partial p_i} + \frac{\partial}{\partial p_i} \left(-\frac{\partial H}{\partial q_i} \right) \right] = \sum_{i=1}^n \left[\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right] = 0 \quad (144)$$

by equality of mixed partials ($H \in C^2$) — note the mechanism: the *minus sign* in Hamilton's equations pairs each expansion rate $\partial \dot{q}_i / \partial q_i$ with an equal and opposite contraction rate $\partial \dot{p}_i / \partial p_i$. For a region advected by a flow, the rate of change of volume is the flux of the velocity field through its boundary [the transport identity; stated — it is the change-of-variables formula for the flow map, differentiated at t], which by the divergence theorem is the integral of the divergence:

$$\frac{d}{dt} \text{vol}(\Omega_t) = \oint_{\partial \Omega_t} X \cdot \vec{n} \, dA = \int_{\Omega_t} \nabla \cdot X \, dV = 0. \quad (145)$$

□

Remark 9.2 (What Liouville forbids and what breaks it). Volume preservation forbids *attractors*: no open set of initial conditions in a Hamiltonian system can funnel into a point or a small cycle, because its volume would have to shrink. Contrast the damped oscillator of the companion document: adding $-b\dot{q}$ makes $\dot{p} = -kq - \frac{b}{m}p$, the divergence becomes $-b/m < 0$, and every region contracts exponentially onto the rest point — dissipation is, precisely, the failure of Liouville. The theorem is also the opening axiom of statistical mechanics: it is what allows a probability density on phase space to be transported like an incompressible fluid.

10 Canonical Transformations and Hamilton–Jacobi Theory

The story of this document has been a story about changes of variables. Theorem 2.11 established that the Lagrangian framework survives *any* smooth invertible change of the coordinates $\vec{q}$; normal coordinates (§5.5) showed how much a good choice delivers — the double pendulum fell apart into two clocks. But the Hamiltonian framework has twice as many variables, and §8 already met a change of variables that mixes them: the mode variables (ξ_r, π_r) of Example 8.2 are built from the θ 's alone, but nothing in the bracket criterion required that. This section asks the question that example left open — *which* transformations of phase space preserve Hamiltonian mechanics — proves the criterion the example used, and then pushes the freedom to its logical end: a transformation so well adapted that the new Hamiltonian is *zero* and the new variables do not move at all. The condition for its generating function is a single partial differential equation, the *Hamilton–Jacobi equation*, and the function that satisfies it turns out to be an old acquaintance: the action itself.

10.1 Canonical transformations

Consider a smooth, invertible, possibly time-dependent change of phase-space variables

$$Q_i = Q_i(\vec{q}, \vec{p}, t), \quad P_i = P_i(\vec{q}, \vec{p}, t), \quad i = 1, \dots, n. \quad (146)$$

Definition 10.1 (Canonical transformation). The transformation (146) is *canonical* if for every Hamiltonian $H(\vec{q}, \vec{p}, t)$ there exists a function $K(\vec{Q}, \vec{P}, t)$ such that curves satisfying Hamilton's equations for $(\vec{q}, \vec{p}; H)$ are carried exactly to curves satisfying Hamilton's equations for $(\vec{Q}, \vec{P}; K)$:

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i}. \quad (147)$$

The demand “for every H ” is what gives the notion teeth: we are classifying changes of variables that respect the *form* of the theory, not tricks adapted to one system — the same standard Theorem 2.11 set on the Lagrangian side.

Two examples calibrate how large the class is. First, every point transformation $\vec{Q} = \vec{Q}(\vec{q}, t)$ extends to a canonical transformation (with P built from the covariant transformation of momenta); this is Theorem 2.11 read through the Legendre transform, and we will recover it from the generating-function machinery below. Second — and this is what point transformations can never do — the swap

$$Q = p, \quad P = -q \quad (148)$$

is canonical. Check it in full for one degree of freedom: define $K(Q, P, t) \equiv H(q, p, t)$, i.e. $K(Q, P, t) = H(-P, Q, t)$. Then along a solution of the old equations,

$$\dot{Q} = \dot{p} = -\frac{\partial H}{\partial q} = -\frac{\partial H}{\partial q}(-P, Q, t) = \frac{\partial K}{\partial P}, \quad \dot{P} = -\dot{q} = -\frac{\partial H}{\partial p} = -\frac{\partial K}{\partial Q}, \quad (149)$$

where the middle steps used the chain rule on $K(Q, P, t) = H(-P, Q, t)$: $\partial K/\partial P = \partial H/\partial q \cdot (-1)$ and $\partial K/\partial Q = \partial H/\partial p$. The new equations hold with the *same* Hamiltonian, relabeled. Coordinates and momenta are not two kinds of object in this formalism; they are two halves of one state, exchangeable at the cost of a sign — the symmetry that Hamilton’s equations (95) wear on their face.

The bracket criterion, now proved. Example 8.2 used, without proof, the statement that a time-independent transformation is canonical precisely when it reproduces the fundamental brackets. Here is the proof, in the direction the example needed.

Proposition 10.2 (Fundamental brackets $\Rightarrow$ canonical; time-independent case). *Let $Q_i(\vec{q}, \vec{p})$, $P_i(\vec{q}, \vec{p})$ be a smooth invertible time-independent transformation satisfying*

$$\{Q_i, Q_j\} = 0, \quad \{P_i, P_j\} = 0, \quad \{Q_i, P_j\} = \delta_{ij} \quad (\text{all brackets in } \vec{q}, \vec{p}). \quad (150)$$

Then the transformation is canonical with $K(\vec{Q}, \vec{P}, t)$ defined by substitution, $K(\vec{Q}(\vec{q}, \vec{p}), \vec{P}(\vec{q}, \vec{p}), t) = H(\vec{q}, \vec{p}, t)$.

Proof. The engine is Proposition 8.3: along any solution, $\dot{F} = \{F, H\}$ for every phase-space function F (with no $\partial F/\partial t$ term here, since Q_i, P_i are time-independent). Apply it to $F = Q_i$:

$$\dot{Q}_i = \{Q_i, H\}. \quad (151)$$

Now express H through the new variables, $H = K(\vec{Q}(\vec{q}, \vec{p}), \vec{P}(\vec{q}, \vec{p}), t)$, and expand the bracket by the chain rule. Writing it out from the definition — no shortcut, since this is the step that carries the proof:

$$\begin{aligned} \{Q_i, H\} &= \sum_{a=1}^n \left[\frac{\partial Q_i}{\partial q_a} \frac{\partial H}{\partial p_a} - \frac{\partial Q_i}{\partial p_a} \frac{\partial H}{\partial q_a} \right] \\ &= \sum_{a=1}^n \left[\frac{\partial Q_i}{\partial q_a} \sum_{j=1}^n \left(\frac{\partial K}{\partial Q_j} \frac{\partial Q_j}{\partial p_a} + \frac{\partial K}{\partial P_j} \frac{\partial P_j}{\partial p_a} \right) - \frac{\partial Q_i}{\partial p_a} \sum_{j=1}^n \left(\frac{\partial K}{\partial Q_j} \frac{\partial Q_j}{\partial q_a} + \frac{\partial K}{\partial P_j} \frac{\partial P_j}{\partial q_a} \right) \right] \\ &= \sum_{j=1}^n \frac{\partial K}{\partial Q_j} \underbrace{\sum_{a=1}^n \left[\frac{\partial Q_i}{\partial q_a} \frac{\partial Q_j}{\partial p_a} - \frac{\partial Q_i}{\partial p_a} \frac{\partial Q_j}{\partial q_a} \right]}_{=\{Q_i, Q_j\}} + \sum_{j=1}^n \frac{\partial K}{\partial P_j} \underbrace{\sum_{a=1}^n \left[\frac{\partial Q_i}{\partial q_a} \frac{\partial P_j}{\partial p_a} - \frac{\partial Q_i}{\partial p_a} \frac{\partial P_j}{\partial q_a} \right]}_{=\{Q_i, P_j\}}. \end{aligned} \quad (152)$$

The first line is the definition of the bracket; the second applies the ordinary chain rule to $\partial H/\partial p_a$ and $\partial H/\partial q_a$; the third merely regroups the four resulting double sums by which derivative of K they carry — and the coefficients that emerge are exactly the fundamental brackets. By hypothesis (150),

$$\dot{Q}_i = \sum_j \frac{\partial K}{\partial Q_j} \cdot 0 + \sum_j \frac{\partial K}{\partial P_j} \delta_{ij} = \frac{\partial K}{\partial P_i}. \quad (153)$$

The identical computation with $F = P_i$ gives

$$\dot{P}_i = \{P_i, H\} = \sum_j \frac{\partial K}{\partial Q_j} \{P_i, Q_j\} + \sum_j \frac{\partial K}{\partial P_j} \{P_i, P_j\} = \sum_j \frac{\partial K}{\partial Q_j} (-\delta_{ij}) = -\frac{\partial K}{\partial Q_i}, \quad (154)$$

using antisymmetry, $\{P_i, Q_j\} = -\{Q_j, P_i\} = -\delta_{ij}$. Both Hamilton equations hold in the new variables, for arbitrary H . $\square$

This closes the loop opened by Example 8.2: the mode variables of the double pendulum, certified there by their brackets, are now certified canonical by proof. Time-*dependent* transformations need one more idea — the new Hamiltonian acquires a correction term, $K \neq H$ in general — and that idea is the generating function.

10.2 Generating functions

The bracket criterion certifies a transformation after the fact. To *manufacture* canonical transformations — and eventually to search for the best one — we need a handle that produces them. The handle comes from the variational principle of §6.2, Derivation 2: Hamilton's equations for $(\vec{q}, \vec{p}; H)$ say exactly that the phase-space action

$$S[\vec{q}, \vec{p}] = \int_{t_0}^{t_1} \left[\sum_i p_i \dot{q}_i - H(\vec{q}, \vec{p}, t) \right] dt \quad (155)$$

is stationary. Two integrands that differ by a total time derivative have the same interior variational content — so if the old and new integrands differ by one, the two sets of Hamilton equations must select the same curves. Making this precise is the next proposition; the care it takes is instructive.

Proposition 10.3 (Equivalence up to a total derivative). *Let (146) be a smooth invertible transformation, and suppose there exist smooth functions $K(\vec{Q}, \vec{P}, t)$ and F (of the phase-space variables and t) such that along every smooth phase-space curve,*

$$\sum_i p_i \dot{q}_i - H = \sum_i P_i \dot{Q}_i - K + \frac{dF}{dt}. \quad (156)$$

Then a curve satisfies Hamilton's equations for $(\vec{q}, \vec{p}; H)$ if and only if its image satisfies Hamilton's equations for $(\vec{Q}, \vec{P}; K)$.

Proof. Integrate (156) along any curve from t_0 to t_1 :

$$S[\vec{q}, \vec{p}] = S'[\vec{Q}, \vec{P}] + F|_{t_1} - F|_{t_0}, \quad (157)$$

where S' is the phase-space action built from $(\vec{Q}, \vec{P}; K)$. Now restrict attention to *compactly supported* variations: variations $(\delta\vec{q}, \delta\vec{p})$ that vanish identically on neighborhoods of t_0 and t_1 (the bump functions of Appendix B are of exactly this kind). Three observations, each doing one job:

(i) *The boundary term drops out.* A compactly supported variation leaves the curve unchanged near the endpoints, so $F|_{t_0}$ and $F|_{t_1}$ — functions of the curve’s endpoint data — are unvaried. Varying (157),

$$\delta S = \delta S' \quad (\text{compactly supported variations}). \quad (158)$$

(ii) *Compact support is preserved by the transformation.* Since (146) is smooth and invertible, a variation of $(\vec{q}, \vec{p})$ vanishing near the endpoints induces a variation of $(\vec{Q}, \vec{P})$ vanishing near the endpoints, and conversely. So “compactly supported” means the same thing on both sides of (158).

(iii) *Stationarity under compactly supported variations is exactly Hamilton’s equations.* Rerun Derivation 2 of §6.2 with compactly supported variations: the $\vec{p}$ -variation needs no boundary condition at all, and the $\vec{q}$ -variation’s single integration by parts produces a boundary term $[p_i \delta q_i]_{t_0}^{t_1}$ that vanishes because δq_i is identically zero near the ends. The fundamental lemma (Appendix B) requires only compactly supported test functions, so it applies verbatim, and delivers $\dot{q}_i = \partial H / \partial p_i$, $\dot{p}_i = -\partial H / \partial q_i$. The same argument runs for $(\vec{Q}, \vec{P}; K)$.

Combining: the curve solves the old equations $\iff \delta S = 0$ for all compactly supported variations $\iff \delta S' = 0$ for all compactly supported variations $\iff$ the image solves the new equations. $\square$

Remark 10.4 (Why the usual endpoint phrasing is delicate). The textbook version of this argument pins $\vec{q}$ at the endpoints, pins $\vec{Q}$ at the endpoints, and declares both boundary terms dead. But those are two different conditions: when $\vec{Q}$ mixes coordinates and momenta, $\delta \vec{q}(t_0) = 0$ does *not* force $\delta \vec{Q}(t_0) = 0$. The variation of the boundary term is $\sum_i (p_i \delta q_i - P_i \delta Q_i)|_{t_0}^{t_1}$, and killing only its first half leaves the argument with a gap. Compactly supported variations close the gap wholesale: they kill every possible boundary term at once, and they are all that the fundamental lemma ever needed. The equations of motion are a statement about the *interior* of the time interval; it is fitting that the clean proof never touches the ends.

Type-2 generating functions. Proposition 10.3 converts the search for canonical transformations into the search for identities of the form (156) — and those can be manufactured from a single scalar function. Among the standard choices of independent variables we develop first the one Jacobi found most useful — a function of the *old coordinates and new momenta* — and then the remaining three, each in full.

Proposition 10.5 (The type-2 construction). *Let $F_2(\vec{q}, \vec{P}, t)$ be smooth on some domain, with*

$$\det\left(\frac{\partial^2 F_2}{\partial q_i \partial P_j}\right) \neq 0. \quad (159)$$

Define the transformation implicitly by

$$p_i = \frac{\partial F_2}{\partial q_i}(\vec{q}, \vec{P}, t), \quad Q_i = \frac{\partial F_2}{\partial P_i}(\vec{q}, \vec{P}, t), \quad K = H + \frac{\partial F_2}{\partial t}. \quad (160)$$

Then: (i) the first relation locally determines $\vec{P}(\vec{q}, \vec{p}, t)$, so (160) defines a genuine transformation $(\vec{q}, \vec{p}) \mapsto (\vec{Q}, \vec{P})$; and (ii) this transformation is canonical, with new Hamiltonian K .

Proof. (i) Fix t . The map $\vec{P} \mapsto \partial F_2 / \partial \vec{q}$ has Jacobian matrix $\partial^2 F_2 / \partial P_j \partial q_i$, whose determinant is nonzero by (159). The implicit function theorem — invoked exactly as in Appendix D — solves $p_i = \partial F_2 / \partial q_i(\vec{q}, \vec{P}, t)$ locally for $\vec{P}(\vec{q}, \vec{p}, t)$, smoothly. Substituting into the second relation yields $\vec{Q}(\vec{q}, \vec{p}, t)$.

(ii) We verify the identity (156) with

$$F = F_2(\vec{q}, \vec{P}, t) - \sum_i P_i Q_i, \quad (161)$$

by direct differentiation along an arbitrary curve. Slowly, term by term:

$$\begin{aligned} \frac{dF}{dt} &= \sum_i \frac{\partial F_2}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial F_2}{\partial P_i} \dot{P}_i + \frac{\partial F_2}{\partial t} - \sum_i \dot{P}_i Q_i - \sum_i P_i \dot{Q}_i \\ &= \sum_i p_i \dot{q}_i + \sum_i Q_i \dot{P}_i + (K - H) - \sum_i Q_i \dot{P}_i - \sum_i P_i \dot{Q}_i \\ &= \left[\sum_i p_i \dot{q}_i - H \right] - \left[\sum_i P_i \dot{Q}_i - K \right], \end{aligned} \quad (162)$$

where the second line substituted all three defining relations (160) (this is the only place they are used, and all three are used). Rearranged, (162) is (156), so Proposition 10.3 applies. $\square$

Note what the third relation in (160) did: the mismatch between old and new Hamiltonian is exactly the explicit time dependence of the generating function. For time-independent F_2 the Hamiltonian carries over unchanged, $K = H$, recovering the setting of Proposition 10.2; for time-dependent F_2 we have gained the ability to *reshape* H — the crack of daylight that Hamilton–Jacobi theory will widen into a method.

Two calibrations. The choice $F_2 = \sum_i q_i P_i$ gives $p_i = P_i$, $Q_i = q_i$, $K = H$: the identity, with nondegeneracy matrix δ_{ij} . And $F_2 = \sum_i f_i(\vec{q}, t) P_i$ gives $Q_i = f_i(\vec{q}, t)$ with $p_i = \sum_j (\partial f_j / \partial q_i) P_j$ — every point transformation, with the momenta transforming by exactly the covariant rule, and $K = H + \sum_i P_i \partial f_i / \partial t$. Theorem 2.11’s content reappears as the two easiest entries of a much larger catalogue.

Remark 10.6 (Near-identity transformations and the bracket, reunited). Perturb the identity: $F_2 = \sum_i q_i P_i + \epsilon G(\vec{q}, \vec{P}, t)$ with ϵ small. The relations (160) give $p_i = P_i + \epsilon \partial G / \partial q_i$ and $Q_i = q_i + \epsilon \partial G / \partial P_i$, so to first order in ϵ (where $\vec{P}$ may be replaced by $\vec{p}$ inside G),

$$Q_i = q_i + \epsilon \frac{\partial G}{\partial P_i} + O(\epsilon^2), \quad P_i = p_i - \epsilon \frac{\partial G}{\partial q_i} + O(\epsilon^2) \implies \delta q_i = \epsilon \{q_i, G\}, \quad \delta p_i = \epsilon \{p_i, G\}. \quad (163)$$

Infinitesimal canonical transformations are bracket flows: every function G *generates* a one-parameter family of canonical maps, moving phase space along $\{\cdot, G\}$. Read against Proposition 8.3, the case $G = H$, $\epsilon = dt$ says: *time evolution itself is a canonical transformation, generated by the Hamiltonian*. §10.6 will exhibit its generating function in closed form.

The other three types, in full. The type-2 function is one entry in a catalogue of four, indexed by which mixed pair of variables — half old, half new — the scalar function depends on. The catalogue is not a convenience menu; it is forced, and the two easiest transformations in this section already show why. The *identity* has $\vec{Q} = \vec{q}$: old and new coordinates are functionally dependent, so no function of $(\vec{q}, \vec{Q})$ can carry independent information about the transformation, and indeed no type-1 function generates it. The *swap* (148) has $\vec{P} = -\vec{q}$: old coordinates and new momenta are functionally dependent, and no type-2 function generates it — the nondegeneracy condition (159) cannot even be posed, because $(\vec{q}, \vec{P})$ fails to be a coordinate system on the transformation’s graph. Each type covers transformations the others cannot; a complete toolkit needs all four. We now construct the remaining three by the same two-part argument as Proposition 10.5 — an implicit function theorem step and a direct verification of (156) — writing each verification out in full, because the sign pattern is exactly where this subject sheds users.

Proposition 10.7 (The type-1 construction). *Let $F_1(\vec{q}, \vec{Q}, t)$ be smooth with $\det(\partial^2 F_1 / \partial q_i \partial Q_j) \neq 0$. Define*

$$p_i = \frac{\partial F_1}{\partial q_i}(\vec{q}, \vec{Q}, t), \quad P_i = -\frac{\partial F_1}{\partial Q_i}(\vec{q}, \vec{Q}, t), \quad K = H + \frac{\partial F_1}{\partial t}. \quad (164)$$

Then the first relation locally determines $\vec{Q}(\vec{q}, \vec{p}, t)$, the second then evaluates $\vec{P}$, and the resulting transformation is canonical with new Hamiltonian K .

Proof. Solvability. At fixed t , the map $\vec{Q} \mapsto \partial F_1 / \partial \vec{Q}$ has Jacobian $(\partial^2 F_1 / \partial Q_j \partial q_i)$, invertible by hypothesis; the implicit function theorem solves $p_i = \partial F_1 / \partial q_i(\vec{q}, \vec{Q}, t)$ locally for $\vec{Q}(\vec{q}, \vec{p}, t)$, smoothly.

Verification. Here the generating function enters the identity (156) bare: take $F = F_1$. Along an arbitrary curve,

$$\begin{aligned} \frac{dF_1}{dt} &= \sum_i \frac{\partial F_1}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial F_1}{\partial Q_i} \dot{Q}_i + \frac{\partial F_1}{\partial t} \\ &= \sum_i p_i \dot{q}_i + \sum_i (-P_i) \dot{Q}_i + (K - H) = \left[\sum_i p_i \dot{q}_i - H \right] - \left[\sum_i P_i \dot{Q}_i - K \right], \end{aligned} \quad (165)$$

substituting all three relations (164) in the second step. This is (156) rearranged; Proposition 10.3 finishes. Note where the minus sign in (164) comes from: F_1 needs no $\sum P_i Q_i$ correction (compare Proposition 10.5), so the $\dot{Q}_i$ coefficient of dF_1/dt must *itself* be $-P_i$ for the identity to close. In the type-2 case that job was done by the subtracted $\sum P_i Q_i$; here it is done by a sign in the definition. Same identity, two ways to produce the same term. $\square$

Proposition 10.8 (The type-3 and type-4 constructions). *Let $F_3(\vec{p}, \vec{Q}, t)$ be smooth with $\det(\partial^2 F_3 / \partial p_i \partial Q_j) \neq 0$, and define*

$$q_i = -\frac{\partial F_3}{\partial p_i}, \quad P_i = -\frac{\partial F_3}{\partial Q_i}, \quad K = H + \frac{\partial F_3}{\partial t}. \quad (166)$$

Let $F_4(\vec{p}, \vec{P}, t)$ be smooth with $\det(\partial^2 F_4 / \partial p_i \partial P_j) \neq 0$, and define

$$q_i = -\frac{\partial F_4}{\partial p_i}, \quad Q_i = \frac{\partial F_4}{\partial P_i}, \quad K = H + \frac{\partial F_4}{\partial t}. \quad (167)$$

Each defines, locally, a canonical transformation with new Hamiltonian K .

Proof. Solvability, both cases, one sentence each. For F_3 : the map $\vec{Q} \mapsto -\partial F_3 / \partial \vec{p}$ has Jacobian $-(\partial^2 F_3 / \partial Q_j \partial p_i)$, invertible, so $q_i = -\partial F_3 / \partial p_i(\vec{p}, \vec{Q}, t)$ solves for $\vec{Q}(\vec{q}, \vec{p}, t)$. For F_4 : the map $\vec{P} \mapsto -\partial F_4 / \partial \vec{p}$ has Jacobian $-(\partial^2 F_4 / \partial P_j \partial p_i)$, invertible, so the first relation solves for $\vec{P}(\vec{q}, \vec{p}, t)$.

Verification for F_3 . The correction now runs on the *old* pair: take

$$F = F_3(\vec{p}, \vec{Q}, t) + \sum_i p_i q_i. \quad (168)$$

Differentiate along an arbitrary curve, term by term, substituting (166) where marked:

$$\begin{aligned} \frac{dF}{dt} &= \sum_i \frac{\partial F_3}{\partial p_i} \dot{p}_i + \sum_i \frac{\partial F_3}{\partial Q_i} \dot{Q}_i + \frac{\partial F_3}{\partial t} + \sum_i \dot{p}_i q_i + \sum_i p_i \dot{q}_i \\ &= \sum_i (-q_i) \dot{p}_i + \sum_i (-P_i) \dot{Q}_i + (K - H) + \sum_i q_i \dot{p}_i + \sum_i p_i \dot{q}_i \\ &= \left[\sum_i p_i \dot{q}_i - H \right] - \left[\sum_i P_i \dot{Q}_i - K \right] : \end{aligned} \quad (169)$$

the $q_i \dot{p}_i$ terms cancel in pairs — the added $\sum p_i q_i$ existed to produce exactly that cancellation — and the survivors assemble into (156).

Verification for F_4 . Both pairs need correcting: take

$$F = F_4(\vec{p}, \vec{P}, t) + \sum_i p_i q_i - \sum_i P_i Q_i. \quad (170)$$

Again in full:

$$\begin{aligned} \frac{dF}{dt} &= \sum_i \frac{\partial F_4}{\partial p_i} \dot{p}_i + \sum_i \frac{\partial F_4}{\partial P_i} \dot{P}_i + \frac{\partial F_4}{\partial t} + \sum_i (\dot{p}_i q_i + p_i \dot{q}_i) - \sum_i (\dot{P}_i Q_i + P_i \dot{Q}_i) \\ &= \sum_i (-q_i) \dot{p}_i + \sum_i Q_i \dot{P}_i + (K - H) + \sum_i q_i \dot{p}_i + \sum_i p_i \dot{q}_i - \sum_i Q_i \dot{P}_i - \sum_i P_i \dot{Q}_i \\ &= \left[\sum_i p_i \dot{q}_i - H \right] - \left[\sum_i P_i \dot{Q}_i - K \right], \end{aligned} \quad (171)$$

with two cancellations this time ($q_i \dot{p}_i$ against its partner, $Q_i \dot{P}_i$ against its partner) — each correction term neutralizing the derivative that lands on the “wrong” member of its pair. Proposition 10.3 applies in both cases. $\square$

Remark 10.9 (The catalogue, assembled). Side by side, with each type’s nondegeneracy condition:

type	arguments	defining relations	nondegeneracy
F_1	$(\vec{q}, \vec{Q}, t)$	$p_i = \partial F_1 / \partial q_i, \quad P_i = -\partial F_1 / \partial Q_i$	$\det(\partial^2 F_1 / \partial q \partial Q) \neq 0$
F_2	$(\vec{q}, \vec{P}, t)$	$p_i = \partial F_2 / \partial q_i, \quad Q_i = \partial F_2 / \partial P_i$	$\det(\partial^2 F_2 / \partial q \partial P) \neq 0$
F_3	$(\vec{p}, \vec{Q}, t)$	$q_i = -\partial F_3 / \partial p_i, \quad P_i = -\partial F_3 / \partial Q_i$	$\det(\partial^2 F_3 / \partial p \partial Q) \neq 0$
F_4	$(\vec{p}, \vec{P}, t)$	$q_i = -\partial F_4 / \partial p_i, \quad Q_i = \partial F_4 / \partial P_i$	$\det(\partial^2 F_4 / \partial p \partial P) \neq 0$

In every row, $K = H + \partial F / \partial t$. The sign pattern is not arbitrary but algorithmic: a derivative with respect to a *coordinate*-type argument returns the conjugate momentum with a + if the argument is old (p from q) and, wherever the $P dQ$ side of the identity must be produced directly rather than through a correction term, with a –; a derivative with respect to a *momentum*-type argument returns the conjugate coordinate, with the sign again fixed by which correction terms were subtracted. The four types are related pairwise by Legendre transforms in the variables being swapped — e.g. $F_1 = F_2 - \sum_i P_i Q_i$ when both exist, with the equivalence verified by comparing (164) and (160) — the same construction that carried L to H in §6.1, here carrying one generating description to another. And a caution, to keep the hypotheses complete: the four *pure* types do not exhaust the possibilities. Nothing prevents a transformation from admitting, say, (q_1, P_2) but neither $(\vec{q}, \vec{Q})$ nor $(\vec{q}, \vec{P})$ as independent variables; mixed per-index choices are then required, and they work by the same argument [stated; the general statement is that locally *some* choice of n old and n new variables is independent, but which one is a property of the transformation, not a convention].

The catalogue in action. Two worked examples, chosen because each defeats one type and showcases another.

The swap, by F_1 . Take $F_1 = \sum_i q_i Q_i$. The relations (164) give, walking through the signs,

$$p_i = \frac{\partial F_1}{\partial q_i} = Q_i, \quad P_i = -\frac{\partial F_1}{\partial Q_i} = -q_i, \quad K = H : \quad (172)$$

exactly the swap (148) that opened this section, with its awkward sign revealed as the type-1 minus sign, and nondegeneracy matrix $\partial^2 F_1 / \partial q_i \partial Q_j = \delta_{ij}$. The identity and the swap now sit symmetrically: the simplest F_2 generates the one, the simplest F_1 generates the other, and neither transformation is reachable from the other's type.

The oscillator, by one lucky guess. Here is what a well-chosen canonical transformation can do *without* any partial differential equation — the trick that Hamilton–Jacobi theory will shortly systematize. For $H = p^2/2m + \frac{1}{2}m\omega^2 q^2$, try

$$F_1(q, Q) = \frac{m\omega q^2}{2} \cot Q. \quad (173)$$

Grind the relations (164), slowly:

$$p = \frac{\partial F_1}{\partial q} = m\omega q \cot Q, \quad P = -\frac{\partial F_1}{\partial Q} = -\frac{m\omega q^2}{2} \cdot \left(-\frac{1}{\sin^2 Q}\right) = \frac{m\omega q^2}{2 \sin^2 Q}. \quad (174)$$

Invert for the old variables: the second relation gives $q^2 = (2P/m\omega) \sin^2 Q$, so

$$q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = m\omega \cot Q \sqrt{\frac{2P}{m\omega}} \sin Q = \sqrt{2m\omega P} \cos Q, \quad (175)$$

taking the positive root (a branch choice of the same kind as (197), and resolved the same way: the final formulas continue smoothly through it). Now substitute into H and watch:

$$H = \frac{(\sqrt{2m\omega P} \cos Q)^2}{2m} + \frac{m\omega^2}{2} \frac{2P}{m\omega} \sin^2 Q = \omega P \cos^2 Q + \omega P \sin^2 Q = \omega P. \quad (176)$$

The coordinate Q has vanished from the Hamiltonian entirely: in the new variables the oscillator is a *free* system with a cyclic coordinate (Proposition 2.5, meeting us again on the far side of the formalism). Since F_1 is time-independent, $K = H = \omega P$, and the new equations of motion solve themselves by inspection:

$$\dot{P} = -\frac{\partial K}{\partial Q} = 0, \quad \dot{Q} = \frac{\partial K}{\partial P} = \omega \implies P = \text{const} = \frac{E}{\omega}, \quad Q = \omega t + Q_0. \quad (177)$$

Substituting back through (175),

$$q(t) = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + Q_0), \quad (178)$$

in exact agreement with the solution that §10.5 will reach by systematic search (Eq. (202), under $Q_0 = \omega\beta$) — and with the constant new momentum $P = E/\omega$ recognizable on sight: it is the *action variable*, the enclosed-area-over- 2π of §7.1, making its second unforced appearance (Remark 10.15 names the general construction). The nondegeneracy hypothesis, checked rather than assumed: $\partial^2 F_1 / \partial q \partial Q = -m\omega q / \sin^2 Q$, nonzero away from $q = 0$ — the transformation is honest on each half-line and degenerates at the turning-phase instants — a degeneracy by now familiar from trigonometric charts. One closing observation, which is the hinge of this section: the obvious question — *where did (173) come from?* — has no answer inside this example. It is a guess, verified after the fact, and the Hamilton–Jacobi equation of §10.3 is precisely the machine that replaces the guess with a search. (§10.6 will add a postscript: the dynamics itself manufactures a function of exactly this shape.)

10.3 The Hamilton–Jacobi equation

Now push the freedom to its limit. The best coordinates the double pendulum admitted made H a sum of two independent pieces; the best coordinates *imaginable* would make H vanish identically, so that

$$\dot{Q}_i = \frac{\partial K}{\partial P_i} = 0, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i} = 0 : \quad (179)$$

every new variable a constant of the motion, the entire dynamics transformed away. Nothing in §10.2 forbids the attempt. Take a type-2 generating function — rename it $S(\vec{q}, \vec{P}, t)$, in anticipation — and impose $K = 0$ in (160):

$$0 = K = H\left(\vec{q}, \frac{\partial S}{\partial \vec{q}}, t\right) + \frac{\partial S}{\partial t}, \quad (180)$$

where $\partial S/\partial \vec{q}$ replaced $\vec{p}$ by the first defining relation. The demand on S is therefore

$$\boxed{H\left(q_1, \dots, q_n, \frac{\partial S}{\partial q_1}, \dots, \frac{\partial S}{\partial q_n}, t\right) + \frac{\partial S}{\partial t} = 0} \quad (181)$$

— the *Hamilton–Jacobi equation*: a single first-order partial differential equation for a single scalar function of $n + 1$ variables, generically nonlinear (the momenta enter H quadratically already for a free particle).

Pause on the count, because it looks too good. Hamilton’s equations are $2n$ coupled ordinary differential equations; (181) is *one* equation. The compression is real but not free, and the fine print is the notion the whole method rests on: we do not need the general solution of (181) — we need one solution *per family*, with enough parameters that the second relation of (160) can be inverted.

Definition 10.10 (Complete integral). A *complete integral* of (181) is a solution $S(\vec{q}; \alpha, t)$ depending smoothly on n parameters $\alpha_1, \dots, \alpha_n$ and satisfying, on its domain,

$$\det\left(\frac{\partial^2 S}{\partial q_i \partial \alpha_j}\right) \neq 0. \quad (182)$$

The parameters are the new momenta of the construction: $P_i = \alpha_i$, constant because $K = 0$. Condition (182) is (159) verbatim, and it quietly enforces everything the informal phrase “*n independent, non-additive constants*” is meant to convey. In particular, since only derivatives of S enter (181), $S + c$ is a solution whenever S is; an additive constant is always available and never counts. Suppose one of the parameters, say α_n , entered only additively, $S = \tilde{S}(\vec{q}; \alpha_1, \dots, \alpha_{n-1}, t) + \alpha_n$. Then $\partial^2 S/\partial q_i \partial \alpha_n = 0$ for every i : an entire column of the matrix in (182) vanishes, and the determinant with it. The nondegeneracy condition is not a technical afterthought bolted onto the definition; it *is* the definition, with the folklore exclusions as corollaries.

10.4 Jacobi’s theorem, by two routes

Here is the payoff, stated with its hypotheses in full view.

Theorem 10.11 (Jacobi). Let $H(\vec{q}, \vec{p}, t)$ be C^1 and let $S(\vec{q}; \alpha, t)$ be a C^2 complete integral of the Hamilton–Jacobi equation (181) on a domain where (182) holds. For each fixed (α, β) , the relations

$$\beta_i = \frac{\partial S}{\partial \alpha_i}(\vec{q}; \alpha, t), \quad p_i = \frac{\partial S}{\partial q_i}(\vec{q}; \alpha, t) \quad (183)$$

determine, locally in t , a curve $(\vec{q}(t), \vec{p}(t))$ — the first relation implicitly, the second by evaluation — and this curve satisfies Hamilton’s equations. Conversely, every solution of Hamilton’s equations passing through the domain arises this way, for suitable constants (α, β) .

We give two proofs of the forward direction. The first travels through the machinery of §§10.2–10.3 and shows *why* the theorem was inevitable; the second uses nothing but differentiation and shows the hypotheses doing their work line by line. A result this central deserves both.

Proof (Route 1: through the canonical machinery). S satisfies the nondegeneracy condition (159), so by Proposition 10.5 it generates a canonical transformation $(\vec{q}, \vec{p}) \mapsto (\vec{Q}, \vec{P}) = (\beta, \alpha)$ with new Hamiltonian $K = H + \partial S / \partial t$ — which vanishes, because S solves (181). In the new variables Hamilton’s equations read $\dot{\beta}_i = \partial K / \partial \alpha_i = 0$ and $\dot{\alpha}_i = -\partial K / \partial \beta_i = 0$: the motion, viewed through S , is standing still. A trajectory is therefore specified by the $2n$ constants (α, β) , and transforming back — i.e. solving (183) for $(\vec{q}(t), \vec{p}(t))$, which part (i) of Proposition 10.5 licenses — produces a solution of the original equations, by Proposition 10.3 read right to left. $\square$

Proof (Route 2: direct verification). First, the curve exists: for fixed (α, β) , the system $\beta_i = \partial S / \partial \alpha_i(\vec{q}; \alpha, t)$ has Jacobian in $\vec{q}$ equal to $(\partial^2 S / \partial \alpha_i \partial q_j)$, invertible by (182); the implicit function theorem gives a unique smooth local solution $\vec{q}(t)$, and we set $p_i(t) = \partial S / \partial q_i(\vec{q}(t); \alpha, t)$.

First Hamilton equation. Differentiate the identity $\beta_i = \partial S / \partial \alpha_i(\vec{q}(t); \alpha, t)$ with respect to t (the left side is constant):

$$0 = \frac{\partial^2 S}{\partial t \partial \alpha_i} + \sum_{j=1}^n \frac{\partial^2 S}{\partial q_j \partial \alpha_i} \dot{q}_j. \quad (184)$$

Separately, S satisfies (181) *identically* in $(\vec{q}, \alpha, t)$ — the parameters are honest variables of the solution — so we may differentiate (181) with respect to α_i at fixed $(\vec{q}, t)$. The Hamiltonian depends on α_i only through its momentum slots $p_j = \partial S / \partial q_j$:

$$0 = \frac{\partial^2 S}{\partial \alpha_i \partial t} + \sum_{j=1}^n \frac{\partial H}{\partial p_j} \left(\vec{q}, \frac{\partial S}{\partial \vec{q}}, t \right) \frac{\partial^2 S}{\partial \alpha_i \partial q_j}. \quad (185)$$

Subtract (185) from (184); the mixed time derivatives cancel ($S \in C^2$, so the order of differentiation is immaterial — this is where that hypothesis works), leaving

$$\sum_{j=1}^n \frac{\partial^2 S}{\partial \alpha_i \partial q_j} \left[\dot{q}_j - \frac{\partial H}{\partial p_j} \right] = 0 \quad (i = 1, \dots, n). \quad (186)$$

This is a square linear system with invertible coefficient matrix (182) acting on the bracketed vector; the only solution is the zero vector:

$$\dot{q}_j = \frac{\partial H}{\partial p_j} \quad (j = 1, \dots, n). \quad (187)$$

Second Hamilton equation. Differentiate $p_i(t) = \partial S / \partial q_i(\vec{q}(t); \alpha, t)$ in t :

$$\dot{p}_i = \frac{\partial^2 S}{\partial t \partial q_i} + \sum_{j=1}^n \frac{\partial^2 S}{\partial q_j \partial q_i} \dot{q}_j. \quad (188)$$

Differentiate (181) with respect to q_i — now H depends on q_i both directly and through every momentum slot:

$$0 = \frac{\partial^2 S}{\partial q_i \partial t} + \frac{\partial H}{\partial q_i} + \sum_{j=1}^n \frac{\partial H}{\partial p_j} \frac{\partial^2 S}{\partial q_i \partial q_j}. \quad (189)$$

Solve (189) for $\partial^2 S / \partial q_i \partial t$, insert into (188), and use (187) to identify $\dot{q}_j = \partial H / \partial p_j$:

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} - \sum_j \frac{\partial H}{\partial p_j} \frac{\partial^2 S}{\partial q_i \partial q_j} + \sum_j \frac{\partial^2 S}{\partial q_j \partial q_i} \frac{\partial H}{\partial p_j} = -\frac{\partial H}{\partial q_i}, \quad (190)$$

the two sums cancelling by equality of mixed partials once more.

Converse (completeness of the recipe). Let $(\bar{q}(t), \bar{p}(t))$ be any solution with data $(\bar{q}^0, \bar{p}^0)$ at time t_0 inside the domain. Consider the system

$$p_i^0 = \frac{\partial S}{\partial q_i}(\bar{q}^0; \alpha, t_0) \quad (i = 1, \dots, n) \quad (191)$$

as equations for α : its Jacobian is $(\partial^2 S / \partial \alpha_j \partial q_i)$ — the transpose of the matrix in (182), hence also invertible — so the implicit function theorem produces (locally) a parameter vector α . Set $\beta_i = \partial S / \partial \alpha_i(\bar{q}^0; \alpha, t_0)$. The trajectory that the forward direction builds from this (α, β) passes through $(\bar{q}^0, \bar{p}^0)$ at t_0 and solves Hamilton's equations; so does $(\bar{q}, \bar{p})$; and solutions of the smooth first-order system through a given point at a given time are unique [existence and uniqueness for the smooth first-order system; stated, as in §6]. The two coincide. $\square$

Remark 10.12 (What each route certifies). Route 1 is short because §10.2 did its work in advance; it explains the theorem — constants appear because the transformed dynamics is trivial. Route 2 is longer and self-contained; it locates every hypothesis — C^2 licenses the two exchanges of mixed partials, and nondegeneracy works three separate times (solvability for $\bar{q}(t)$, the linear-algebra step to (187), and the converse's solvability for α). The two proofs agreeing is not redundancy; it is the certificate that the machinery of generating functions and the bare calculus are describing the same mathematics.

Remark 10.13 (A particular family suffices). Nothing above asked for the general solution of the partial differential equation (181) — a hopeless demand — only for *some* n -parameter family satisfying a determinant condition. This is the method's entire pragmatics: one lucky guess with enough knobs, and Theorem 10.11 converts it into every trajectory. The next subsection makes the guess for the oscillator and watches the conversion happen.

10.5 The oscillator, one more time

Warm-up: the free particle. $H = p^2/2m$. The Hamilton–Jacobi equation reads $\frac{1}{2m}(\partial S / \partial q)^2 + \partial S / \partial t = 0$. Try $S = \alpha q - f(t)$: then $\alpha^2/2m = f'(t)$, so

$$S(q; \alpha, t) = \alpha q - \frac{\alpha^2}{2m} t. \quad (192)$$

Check completeness: $\partial^2 S / \partial q \partial \alpha = 1 \neq 0$. Turn Jacobi's crank:

$$\beta = \frac{\partial S}{\partial \alpha} = q - \frac{\alpha}{m} t \implies q(t) = \beta + \frac{\alpha}{m} t, \quad p = \frac{\partial S}{\partial q} = \alpha. \quad (193)$$

Uniform motion, with α the (conserved) momentum and β the initial position — the constants of Theorem 10.11 arriving pre-labeled with physical meaning.

Separating the time. For any time-independent Hamiltonian the same first move works. Insert the *ansatz*

$$S(\vec{q}; \alpha, t) = W(\vec{q}; \alpha) - Et \quad (194)$$

into (181): the time derivative contributes the constant $-E$, every other term is time-independent, and the equation collapses to

$$H\left(\vec{q}, \frac{\partial W}{\partial \vec{q}}\right) = E \quad (195)$$

— the *reduced* (time-independent) Hamilton–Jacobi equation, with the separation constant E available as one of the parameters α_i . To be clear about the logic: (194) is not claimed to exhaust the solutions of (181); it is a guess, and by the previous remark a guess is all the theorem asks. If solving (195) yields a family with the right count and a nonvanishing determinant, Theorem 10.11 does not care how the family was found. Equation (195) also wears its meaning openly: the momentum field $\vec{p} = \partial W / \partial \vec{q}$ is constrained to the energy surface $H = E$ — the same surfaces that organized the phase portraits of §7.1.

The oscillator. $H = p^2/2m + \frac{1}{2}m\omega^2 q^2$, one degree of freedom, so the single parameter can be taken to be E itself ($\alpha \equiv E$). The reduced equation

$$\frac{1}{2m} \left(\frac{dW}{dq} \right)^2 + \frac{1}{2} m \omega^2 q^2 = E \quad (196)$$

is algebraic in W' :

$$\frac{dW}{dq} = \pm \sqrt{2mE - m^2 \omega^2 q^2}. \quad (197)$$

Here the derivation slows down, because a sign and a domain have just appeared. The radicand is nonnegative only for $|q| \leq A$, where

$$A \equiv \sqrt{\frac{2E}{m\omega^2}} \quad (198)$$

— the classical turning points, at which $W' = 0$. Choose the $+$ branch on the open interval $-A < q < A$; the $-$ branch is the return sweep, and we will see in a moment that the final answer accounts for it without being asked. On the chosen branch,

$$W(q; E) = \int^q \sqrt{2mE - m^2 \omega^2 s^2} \, ds, \quad (199)$$

with the lower limit left unwritten: changing it shifts W by a constant, which the completeness discussion has already told us is invisible.

Now the labor-saving step. Jacobi’s recipe wants $\beta = \partial S / \partial E$, and differentiating (199) *under the integral sign* (licensed by Appendix A; the integrand is smooth in (s, E) on the open branch) gives

$$\beta = \frac{\partial S}{\partial E} = \frac{\partial W}{\partial E} - t = \int^q \frac{m \, ds}{\sqrt{2mE - m^2 \omega^2 s^2}} - t, \quad (200)$$

— and the integral that remains is *easier* than (199): the recipe never needed W itself, only its E -derivative. Factor the radicand, $\sqrt{2mE - m^2 \omega^2 s^2} = m\omega \sqrt{A^2 - s^2}$:

$$\beta + t = \frac{1}{\omega} \int^q \frac{ds}{\sqrt{A^2 - s^2}} = \frac{1}{\omega} \arcsin \frac{q}{A} + \text{const}, \quad (201)$$

absorbing the constant into β (it shifts the origin of time, which is what β is for). The arcsin is the principal branch, valued in $[-\pi/2, \pi/2]$ — a single sweep from $-A$ to A , faithful to the single branch of (197) we integrated. Inverting,

$$\boxed{q(t) = A \sin(\omega(t + \beta)), \quad A = \sqrt{\frac{2E}{m\omega^2}}} \quad (202)$$

— and the inversion is where the branch bookkeeping resolves itself: arcsin was single-valued, but sin is entire, and (202) happily continues through the turning points and back, executing the — branch of (197) on the return sweep without further instruction. One can check the claim directly: $p = \partial S / \partial q = W'(q) = m\omega\sqrt{A^2 - q^2}$ on the + branch, while (202) gives $m\dot{q} = m\omega A \cos(\omega(t + \beta))$, and the two agree in magnitude with the sign of the cosine selecting the branch — positive on the outbound sweep, negative on the return, switching exactly at $q = \pm A$.

Three closures, in order of increasing reach. (i) *The hypothesis, concretely*: on the open branch,

$$\frac{\partial^2 S}{\partial q \partial E} = \frac{\partial}{\partial E} \sqrt{2mE - m^2\omega^2 q^2} = \frac{m}{\sqrt{2mE - m^2\omega^2 q^2}} \neq 0, \quad (203)$$

finite and nonzero for $|q| < A$ — Definition 10.10 verified rather than assumed, with its failure at the turning points visible in the formula (the very points we excised from the branch). (ii) *Back-substitution*: (202) inserted into both Hamilton equations: $\dot{q} = \omega A \cos(\omega(t + \beta)) = p/m$ ✓, and $\dot{p} = \frac{d}{dt}[m\omega A \cos(\omega(t + \beta))] = -m\omega^2 A \sin(\omega(t + \beta)) = -m\omega^2 q$ ✓. (iii) *Against the series*: (202) is the amplitude–phase form of [Osc. §2.3] with amplitude $A = \sqrt{2E/m\omega^2}$, i.e. $E = \frac{1}{2}m\omega^2 A^2$ — which is precisely the constant total energy computed in [Osc. §2.5]. Three formalisms and two documents meet at one formula.

10.6 The action as the generating function

The Hamilton–Jacobi equation was manufactured abstractly: demand $K = 0$, obtain a PDE, hunt for complete integrals. This subsection shows that the dynamics has been solving that PDE all along — the solution it builds is the action, evaluated on true trajectories and read as a function of where they end.

Setup, with its hypothesis in the open. Fix an initial event $(\vec{q}_0, t_0)$. Suppose that for every endpoint $(\vec{q}, t)$ in some neighborhood there is a *unique* solution of the equations of motion running from $(\vec{q}_0, t_0)$ to $(\vec{q}, t)$, depending smoothly on the endpoint.¹ Define *Hamilton’s principal function*

$$S(\vec{q}, t) \equiv \int_{t_0}^t L(\vec{q}(\tau), \dot{\vec{q}}(\tau)) d\tau \quad \text{along the true trajectory } \vec{q} \text{ from } (\vec{q}_0, t_0) \text{ to } (\vec{q}, t). \quad (204)$$

The same integral that Hamilton’s principle (§3.1) compared across *virtual* paths is here evaluated only on *real* ones and studied as a function of the endpoint. Everything below follows from the two derivatives of S .

¹This is a genuine hypothesis, not a pleasantry. Existence and smooth dependence hold near a given trajectory for t close to t_0 [smooth dependence of ODE solutions on data and parameters; stated — the same import family as the uniqueness theorem used in §6]. Uniqueness of the *boundary-value* problem, unlike the initial-value problem, can genuinely fail at finite times: the oscillator below fails it at $\omega(t - t_0) = \pi$, visibly, in Fig. 4.

Spatial derivative. Vary the endpoint position at fixed time: $\vec{q} \rightarrow \vec{q} + \delta\vec{q}$. The trajectory family shifts, $\vec{q} \rightarrow \vec{q} + \delta\vec{q}$ with $\delta\vec{q}(t_0) = 0$ (the start is nailed down) and $\delta\vec{q}(t) = \delta\vec{q}$. Run the variation exactly as in the proof of Theorem 2.1 — expand, integrate by parts — but keep the boundary term this time:

$$\delta S = \int_{t_0}^t \sum_i \underbrace{\left[\frac{\partial L}{\partial q_i} - \frac{d}{d\tau} \frac{\partial L}{\partial \dot{q}_i} \right]}_{= 0 \text{ on a true trajectory}} \delta \bar{q}_i d\tau + \left[\sum_i \frac{\partial L}{\partial \dot{q}_i} \delta \bar{q}_i \right]_{t_0}^t. \quad (205)$$

The bulk dies because $\vec{q}$ solves Euler–Lagrange — in Hamilton’s principle the *variation* was constrained and the bulk survived to become the equations of motion; here the *path* is constrained to be physical and the roles reverse: the boundary term, which §2.2 killed with pinned endpoints, is now the entire result. At t_0 it vanishes ($\delta\vec{q}(t_0) = 0$); at t it reads $\sum_i p_i \delta q_i$ with $p_i = \partial L / \partial \dot{q}_i$ the terminal momentum. Hence

$$\frac{\partial S}{\partial q_i} = p_i. \quad (206)$$

Time derivative. Extend the endpoint *along the trajectory itself*: comparing S at $(\vec{q}(t+dt), t+dt)$ with S at $(\vec{q}(t), t)$ appends one sliver of integrand to (204),

$$\frac{d}{dt} S(\vec{q}(t), t) = L. \quad (207)$$

But the left side, by the chain rule, is $\partial S / \partial t + \sum_i (\partial S / \partial q_i) \dot{q}_i = \partial S / \partial t + \sum_i p_i \dot{q}_i$, using (206). Therefore

$$\frac{\partial S}{\partial t} = L - \sum_i p_i \dot{q}_i = -H\left(\vec{q}, \frac{\partial S}{\partial \vec{q}}, t\right), \quad (208)$$

recognizing the Legendre transform (92) and eliminating $\vec{p}$ by (206). Assembling (206)–(208) into one statement:

$$\boxed{dS = \sum_i p_i dq_i - H dt, \quad \text{hence} \quad H\left(\vec{q}, \frac{\partial S}{\partial \vec{q}}, t\right) + \frac{\partial S}{\partial t} = 0} \quad (209)$$

— *Hamilton’s principal function solves the Hamilton–Jacobi equation.* The PDE we conjured by demanding $K = 0$ is satisfied by an object the theory already owned.

The motion is its own canonical transformation. Vary instead the *initial* point $\vec{q}_0$ at fixed $(\vec{q}, t, t_0)$: the same computation runs, but now the boundary term survives at t_0 with a minus sign (it is the lower limit) and dies at t :

$$\frac{\partial S}{\partial q_{0,i}} = -p_{0,i}, \quad (210)$$

the initial momentum. Read (206) and (210) together, treating $S(\vec{q}, \vec{q}_0, t)$ as a function of old coordinates $\vec{q}$ and “new coordinates” $\vec{q}_0$: they are exactly the defining relations (164) of a type-1 generating function (Proposition 10.7) — and the transformation it generates, $(\vec{q}, \vec{p}) \mapsto (\vec{q}_0, \vec{p}_0)$, is the map from the current state back to the initial data: *time evolution, run in reverse*. Remark 10.6 promised that time evolution is a canonical transformation and named its infinitesimal generator (H , through the bracket); here is the finite version, with the generating function exhibited in closed form: it is the classical action. The solvability condition for this reading is $\det(\partial^2 S / \partial q_i \partial q_{0,j}) \neq 0$ — watch it in the examples.

Two principal functions in closed form. *Free particle.* The true path from (q_0, t_0) to (q, t) is uniform, $\dot{q} = (q - q_0)/(t - t_0)$, so

$$S = \int_{t_0}^t \frac{m\dot{q}^2}{2} d\tau = \frac{m(q - q_0)^2}{2(t - t_0)}. \quad (211)$$

Verify the Hamilton–Jacobi equation by substitution: $\partial S/\partial q = m(q - q_0)/(t - t_0)$ (which is indeed $m\dot{q} = p$ ✓), and

$$\frac{\partial S}{\partial t} = -\frac{m(q - q_0)^2}{2(t - t_0)^2} = -\frac{1}{2m} \left(\frac{\partial S}{\partial q} \right)^2 \quad \checkmark \quad (212)$$

The cross-derivative is $\partial^2 S/\partial q \partial q_0 = -m/(t - t_0)$: nonzero for all $t > t_0$. The free particle’s endpoint problem never degenerates — one straight line per endpoint pair, always.

Oscillator. The boundary-value solution from (q_0, t_0) to (q, t) , with $T \equiv t - t_0$ and $\sin \omega T \neq 0$, is

$$\bar{q}(\tau) = \frac{q_0 \sin \omega(t - \tau) + q \sin \omega(\tau - t_0)}{\sin \omega T}, \quad (213)$$

checked in one look: it solves $\ddot{\bar{q}} = -\omega^2 \bar{q}$ (each term is a sinusoid of frequency ω in τ), equals q_0 at $\tau = t_0$ and q at $\tau = t$. For the action, integrate by parts once and let the equation of motion clear the bulk — the same maneuver that proved (206), now used as a computational device:

$$S = \int_{t_0}^t \left[\frac{m\dot{\bar{q}}^2}{2} - \frac{m\omega^2 \bar{q}^2}{2} \right] d\tau = \frac{m}{2} \left[\bar{q} \dot{\bar{q}} \right]_{t_0}^t - \frac{m}{2} \int_{t_0}^t \underbrace{\bar{q} (\ddot{\bar{q}} + \omega^2 \bar{q})}_{=0} d\tau = \frac{m}{2} (q \dot{\bar{q}}(t) - q_0 \dot{\bar{q}}(t_0)). \quad (214)$$

Differentiate (213) and evaluate at the two ends:

$$\dot{\bar{q}}(t) = \frac{\omega(q \cos \omega T - q_0)}{\sin \omega T}, \quad \dot{\bar{q}}(t_0) = \frac{\omega(q - q_0 \cos \omega T)}{\sin \omega T}, \quad (215)$$

so that

$$S = \frac{m\omega}{2 \sin \omega T} \left[(q^2 + q_0^2) \cos \omega T - 2q q_0 \right]. \quad (216)$$

Consistency checks, each one line. $\partial S/\partial q = \frac{m\omega}{\sin \omega T} (q \cos \omega T - q_0) = m\dot{\bar{q}}(t)$ ✓, matching (206). The limit $\omega \rightarrow 0$ at fixed T : $\cos \omega T \rightarrow 1$, $\sin \omega T \rightarrow \omega T$, and (216) $\rightarrow m[(q^2 + q_0^2) - 2q q_0]/2T = m(q - q_0)^2/2T$, recovering the free particle (211) — the general formula passing through its special case. And the promised postscript to §10.2: at $q_0 = 0$, (216) reduces to $(m\omega q^2/2) \cot \omega t$ — the *same function* as the lucky guess (173), evaluated at $Q = \omega t$. No coincidence: in the variables of that example the angle advances as $Q = \omega t + Q_0$, the orbits launched from $q = 0$ have $Q_0 = 0$, and both functions store the same information in their q -derivative — the momentum field $p = m\omega q \cot(\omega t)$ of that one family of orbits. The guess of §10.2 was the dynamics’ own principal function, met before it had a name. That (216) satisfies the oscillator’s Hamilton–Jacobi equation is verified symbolically in the figure script (`make_figures_lagrangian.py`) [verified here, by machine].

The cross-derivative of (216) is

$$\frac{\partial^2 S}{\partial q \partial q_0} = -\frac{m\omega}{\sin \omega T} : \quad (217)$$

nonzero whenever defined, but blowing up as $\omega T \rightarrow \pi$ — and at $\omega T = \pi$ the formula (213) fails outright. The geometry behind the failure is in Fig. 4: every trajectory launched from $q_0 = 0$, regardless of amplitude, returns to $q = 0$ at $\omega t = \pi$ (half a period — isochrony again, the same fact

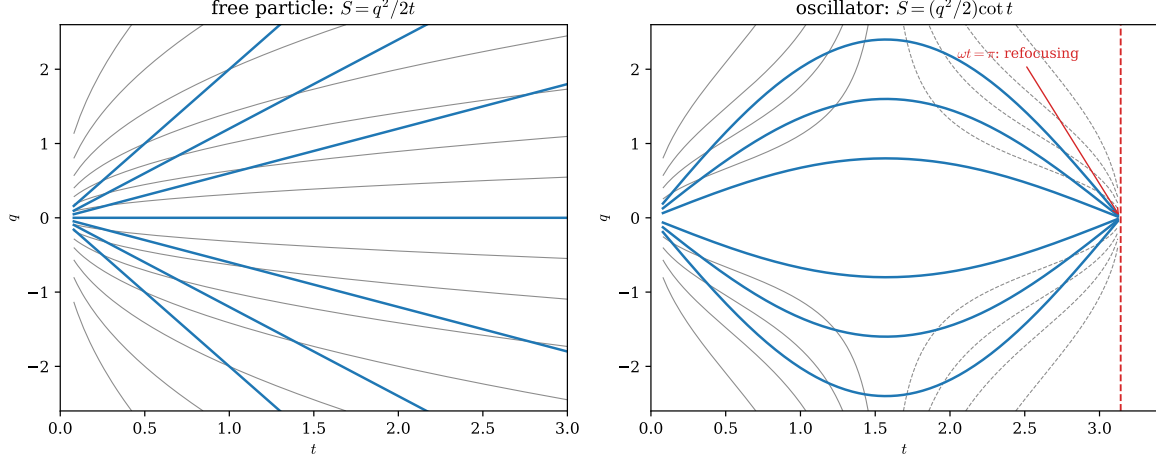


Figure 4: Hamilton's principal function as a field over endpoints (t, q) , launched from $(q_0, t_0) = (0, 0)$; level sets of S (thin) with true trajectories (thick) crossing them. *Left*: free particle, $S = mq^2/2t$ (Eq. (211) with $q_0 = 0$), trajectories $q = vt$; every endpoint with $t > 0$ is reached by exactly one trajectory. *Right*: oscillator, $S = (m\omega q^2/2) \cot \omega t$ (Eq. (216) with $q_0 = 0$), trajectories $q = A \sin \omega t$; all trajectories refocus at $\omega t = \pi$, where the endpoint map degenerates ($\partial^2 S / \partial q \partial q_0 = -m\omega / \sin \omega t \rightarrow \infty$) and the principal function ceases to exist — the failure of the uniqueness hypothesis of §10.6, made visible. Parameters $m = \omega = 1$; both panels drawn from the exact formulas by the figure script, which also verifies both formulas against the Hamilton–Jacobi equation symbolically.

that made the small-angle clock honest in §4). At that instant the endpoint $(0, \pi/\omega)$ is reached by *every* trajectory and the endpoints $(q \neq 0, \pi/\omega)$ by none: the unique-connection hypothesis collapses, and with it the principal function. Such refocusing points are called *conjugate points* (in optics, *caustics*), and they are the honest boundary of this subsection's construction [stated; the systematic theory belongs to the calculus of variations' second-variation chapter].

Remark 10.14 (The quantum pointer, made quantitative). Write a complex field from the principal function, $\psi(q, t) = e^{iS(q, t)/\hbar}$, with $\hbar$ a constant carrying the dimensions of action, and test it against the free Schrödinger operator. Differentiate carefully:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\partial S}{\partial t} \psi, \quad -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial q^2} = \left[\frac{1}{2m} \left(\frac{\partial S}{\partial q} \right)^2 - \frac{i\hbar}{2m} \frac{\partial^2 S}{\partial q^2} \right] \psi, \quad (218)$$

(the first term of the second bracket from the square of $\partial_q S$, the second from the derivative acting twice — one power of $\hbar$ survives). Subtract, carrying a potential term $V\psi$ along on both sides:

$$i\hbar \frac{\partial \psi}{\partial t} - \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial q^2} + V \right] \psi = - \underbrace{\left[\frac{\partial S}{\partial t} + \frac{1}{2m} \left(\frac{\partial S}{\partial q} \right)^2 + V \right]}_{= 0 \text{ by Hamilton–Jacobi}} \psi + \frac{i\hbar}{2m} \frac{\partial^2 S}{\partial q^2} \psi = \frac{i\hbar}{2m} \frac{\partial^2 S}{\partial q^2} \psi. \quad (219)$$

The classical equation annihilates everything of order $\hbar^0$: the Hamilton–Jacobi equation is what the Schrödinger equation becomes when $\hbar$ is negligible against the actions in play, with the residue — one power of $\hbar$, proportional to $\partial_q^2 S$ — repaired by promoting the amplitude of ψ to a second unknown [stated; the systematic expansion is quantum mechanics' WKB approximation]. This is the precise sense in which classical mechanics is the stationary skeleton of a wave theory, and it

dovetails with the path-integral route of the quantum companion: there, each path is weighted by exactly the phase $e^{iS/\hbar}$, and the classical trajectory emerges by stationary phase [QM. §10.3]. The function this section extracted from the endpoint variation is the phase of the quantum propagator.

Remark 10.15 (Beyond: action–angle variables, and the fence). Two directions continue from here; both are pointed at, not entered. *First*, for bounded periodic motion the natural constant is not E but the phase-space area enclosed by the orbit — the *action variable* $I = \oint p dq / 2\pi$ whose first, unforced appearance was the pendulum’s E/ω_0 in §7.1, and whose second was the conserved new momentum $P = E/\omega$ of the oscillator example in §10.2. A complete integral built with I as the new momentum makes the conjugate *angle* advance uniformly, and the pair (I, θ) is the natural language for slow perturbations (adiabatic invariants) and for perturbation theory generally [stated; developed under the name action–angle variables]. *Second*, a fence: Jacobi’s theorem is local, and its global reach is a genuine dynamical question. Complete integrals with global, single-valued W exist exactly when the system possesses n independent conserved quantities in involution — the *integrable* case [stated; the Liouville–Arnold theorem] — and the double pendulum at large amplitude (§11) is the standing reminder that this is the exception, not the rule: no choice of coordinates untangles a chaotic flow, and the Hamilton–Jacobi program, pushed against such a system, fails not by technique but by theorem. What survives universally is the local statement proved here, and the variational identity $dS = \sum_i p_i dq_i - H dt$ that no amount of chaos revokes.

11 Outlook

Three continuations are visible from here; this section only points at them.

Beyond Hamilton–Jacobi. §10 built canonical transformations and pushed them to the $K = 0$ limit. The road continues: action–angle variables (the version of the construction adapted to bounded periodic motion, Remark 10.15), adiabatic invariants, and canonical perturbation theory — the machinery for systems *close* to integrable, and the road on which the failures of integrability (KAM theory, chaos) were first mapped [all stated at pointer level].

The double pendulum at large amplitude. Our §5 lives in the quadratic basin. The exact system (79)–(80) at higher energy is the standard laboratory example of *chaos*: deterministic, energy-conserving (Liouville and H -conservation hold exactly), yet with neighboring initial conditions separating exponentially fast, so that long-term prediction fails in practice while the phase-space structure theorems continue to hold. That these two facts coexist — perfect determinism, practical unpredictability — is a central lesson of twentieth-century mechanics, and the double pendulum is its cheapest demonstration: two rods, two bobs.

Not covered. Rigid-body motion and velocity-dependent potentials (the charged particle’s $L = \frac{1}{2}m\dot{\vec{r}}^2 - q\phi + q\vec{r} \cdot \vec{A}$, the classical seed of gauge coupling) fit the framework built here without modification; non-holonomic constraints (rolling) genuinely require more [all three stated at pointer level].

The bracket becomes a commutator. The remark closing §8 leads directly into quantum mechanics: the Hamiltonian formalism is the classical half of a correspondence in which Poisson brackets become commutators and the phase-space flow becomes unitary evolution. Nothing in Parts I–II has to be unlearned; it has to be *deformed*, with $\hbar$ as the deformation parameter.

A Differentiation Under the Integral Sign

Theorem A.1 (Leibniz rule, the case used in §2.2). *Let $f(x, \varepsilon)$ and $\partial f / \partial \varepsilon$ be continuous on $[a, b] \times [-\delta, \delta]$. Then $\Phi(\varepsilon) = \int_a^b f(x, \varepsilon) dx$ is differentiable and*

$$\Phi'(\varepsilon) = \int_a^b \frac{\partial f}{\partial \varepsilon}(x, \varepsilon) dx. \quad (220)$$

Proof. Fix ε ; for $h \neq 0$ small,

$$\frac{\Phi(\varepsilon + h) - \Phi(\varepsilon)}{h} - \int_a^b \frac{\partial f}{\partial \varepsilon}(x, \varepsilon) dx = \int_a^b \left[\frac{f(x, \varepsilon + h) - f(x, \varepsilon)}{h} - \frac{\partial f}{\partial \varepsilon}(x, \varepsilon) \right] dx. \quad (221)$$

By the mean value theorem in ε , for each x there is ξ_x between ε and $\varepsilon + h$ with

$$\frac{f(x, \varepsilon + h) - f(x, \varepsilon)}{h} = \frac{\partial f}{\partial \varepsilon}(x, \xi_x), \quad (222)$$

so the bracket equals $\partial_\varepsilon f(x, \xi_x) - \partial_\varepsilon f(x, \varepsilon)$ with $|\xi_x - \varepsilon| \leq |h|$. Now the one analytic fact: $\partial_\varepsilon f$ is continuous on the compact set $[a, b] \times [-\delta, \delta]$, hence *uniformly* continuous there. Given $\epsilon' > 0$ choose h small enough that $|\partial_\varepsilon f(x, \xi) - \partial_\varepsilon f(x, \varepsilon)| < \epsilon'$ whenever $|\xi - \varepsilon| \leq |h|$, *uniformly in x* . The integral is then bounded by $(b - a)\epsilon'$, which proves the limit. $\square$

In §2.2 this is applied with $f(x, \varepsilon) = F(x, y + \varepsilon\eta, y' + \varepsilon\eta')$: $F \in C^2$ and $y, \eta \in C^2$ make f and $\partial_\varepsilon f$ continuous on the required compact rectangle.

B The Fundamental Lemma: Full Proof

Lemma B.1. *Let g be continuous on $[a, b]$ and suppose $\int_a^b g \eta dx = 0$ for every $\eta \in C^2([a, b])$ with $\eta(a) = \eta(b) = 0$. Then $g \equiv 0$ on $[a, b]$.*

Proof. Suppose not: $g(x_0) \neq 0$ for some x_0 ; without loss of generality $g(x_0) = 2c > 0$ (else replace g by $-g$, permissible since $-\eta$ is also a test function). By continuity there is a closed interval $[\alpha, \beta] \subset (a, b)$ containing x_0 on which $g \geq c > 0$. Construct the explicit bump

$$\eta(x) = \begin{cases} (x - \alpha)^3 (\beta - x)^3, & x \in [\alpha, \beta], \\ 0, & \text{otherwise.} \end{cases} \quad (223)$$

Admissibility: η vanishes at a and b (its support is inside (a, b)); it is C^2 across the seams $x = \alpha, \beta$ because each factor has a *triple* zero there, so η, η', η'' all vanish from inside, matching the zero function outside (a cubic-order zero leaves two derivatives intact). *Positivity:* $\eta > 0$ on the open interval (α, β) . Then

$$\int_a^b g \eta dx = \int_\alpha^\beta g \eta dx \geq c \int_\alpha^\beta \eta dx > 0, \quad (224)$$

contradicting the hypothesis. Hence $g \equiv 0$. $\square$

(If test functions are required merely continuous or C^1 , the same bump works *a fortiori*; if C^∞ test functions are demanded, replace the cubic factors by the standard $\exp(-1/(x - \alpha)) \exp(-1/(\beta - x))$ bump.)

C Covariance of the Euler–Lagrange Equations: Proof

Setting (Theorem 2.11): $q_i = q_i(\vec{Q}, t)$, a C^2 change of coordinates with invertible Jacobian $J_{ij} = \partial q_i / \partial Q_j$, and $\tilde{L}(\vec{Q}, \dot{\vec{Q}}, t) = L(q(\vec{Q}, t), \dot{q}(\vec{Q}, \dot{\vec{Q}}, t), t)$, where velocities transform by the chain rule

$$\dot{q}_i = \sum_j \frac{\partial q_i}{\partial Q_j} \dot{Q}_j + \frac{\partial q_i}{\partial t}. \quad (225)$$

Proof. Two identities, precisely the (K1)–(K2) of §3.2 with $\vec{r} \rightarrow q$ and $q \rightarrow Q$, both read off (225):

$$(K1) \quad \frac{\partial \dot{q}_i}{\partial \dot{Q}_j} = \frac{\partial q_i}{\partial Q_j}, \quad (K2) \quad \frac{d}{dt} \frac{\partial q_i}{\partial Q_j} = \frac{\partial \dot{q}_i}{\partial Q_j} \quad (226)$$

— (K1) because $\dot{Q}_j$ enters linearly; (K2) because both sides equal $\sum_k \frac{\partial^2 q_i}{\partial Q_k \partial Q_j} \dot{Q}_k + \frac{\partial^2 q_i}{\partial t \partial Q_j}$ (mixed partials of the C^2 map commute). Now compute the Q_j Euler–Lagrange expression of $\tilde{L}$ by the chain rule. Momentum slot:

$$\frac{\partial \tilde{L}}{\partial \dot{Q}_j} = \sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{\partial \dot{q}_i}{\partial \dot{Q}_j} \stackrel{(K1)}{=} \sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{\partial q_i}{\partial Q_j}, \quad (227)$$

$$\frac{d}{dt} \frac{\partial \tilde{L}}{\partial \dot{Q}_j} = \sum_i \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right] \frac{\partial q_i}{\partial Q_j} + \sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{d}{dt} \frac{\partial q_i}{\partial Q_j} \stackrel{(K2)}{=} \sum_i \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right] \frac{\partial q_i}{\partial Q_j} + \sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{\partial \dot{q}_i}{\partial Q_j}. \quad (228)$$

Coordinate slot:

$$\frac{\partial \tilde{L}}{\partial Q_j} = \sum_i \frac{\partial L}{\partial q_i} \frac{\partial q_i}{\partial Q_j} + \sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{\partial \dot{q}_i}{\partial Q_j}. \quad (229)$$

Subtract: the $\partial L / \partial \dot{q}_i \cdot \partial \dot{q}_i / \partial Q_j$ sums cancel, leaving

$$\frac{d}{dt} \frac{\partial \tilde{L}}{\partial \dot{Q}_j} - \frac{\partial \tilde{L}}{\partial Q_j} = \sum_i \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} \right] \frac{\partial q_i}{\partial Q_j}. \quad (230)$$

The new Euler–Lagrange expressions are the old ones contracted with the Jacobian. Since the Jacobian is invertible, the left side vanishes for all j iff the bracket vanishes for all i . $\square$

Equation (230) says more than the theorem: the Euler–Lagrange expression transforms as a *covector* under point transformations — the differential-geometric identity behind the slogan “the equations are tensorial.”

D The Isoperimetric Multiplier Rule: Proof

Setting (Theorem 2.9): y_* stationary for J subject to $\Lambda[y] = L$; y_* not an unconstrained extremal of Λ . Write, for test functions η ,

$$A[\eta] \equiv \delta J[y_*; \eta], \quad B[\eta] \equiv \delta \Lambda[y_*; \eta] \quad (231)$$

(both are integrals of the Euler–Lagrange-type bracket against η , by Moves 1–3 of Theorem 2.1’s proof).

Proof. Since y_* is not an extremal of Λ , the fundamental lemma read backwards supplies a test function η_2 with $B[\eta_2] \neq 0$; normalize $B[\eta_2] = 1$. Fix an arbitrary test function η_1 and consider the two-parameter family

$$y_{\varepsilon_1, \varepsilon_2} = y_* + \varepsilon_1 \eta_1 + \varepsilon_2 \eta_2, \quad \psi(\varepsilon_1, \varepsilon_2) \equiv \Lambda[y_{\varepsilon_1, \varepsilon_2}] - L. \quad (232)$$

Then $\psi(0, 0) = 0$ and $\partial\psi/\partial\varepsilon_2(0, 0) = B[\eta_2] = 1 \neq 0$, so the implicit function theorem yields a C^1 function $\varepsilon_2 = e(\varepsilon_1)$ near 0 with $e(0) = 0$, $\psi(\varepsilon_1, e(\varepsilon_1)) \equiv 0$, and

$$e'(0) = - \left. \frac{\partial\psi/\partial\varepsilon_1}{\partial\psi/\partial\varepsilon_2} \right|_{(0,0)} = -B[\eta_1]. \quad (233)$$

The curve $\varepsilon_1 \mapsto y_{\varepsilon_1, e(\varepsilon_1)}$ is a one-parameter family of *admissible* (constraint-respecting) competitors through y_* ; stationarity of J among admissible curves gives

$$0 = \left. \frac{d}{d\varepsilon_1} \right|_0 J[y_{\varepsilon_1, e(\varepsilon_1)}] = A[\eta_1] + A[\eta_2] e'(0) = A[\eta_1] - A[\eta_2] B[\eta_1]. \quad (234)$$

Set $\lambda \equiv -A[\eta_2]$ — a single constant, fixed once by η_2 and independent of η_1 . The display reads

$$A[\eta_1] + \lambda B[\eta_1] = 0 \quad \text{for every test function } \eta_1, \quad (235)$$

i.e. $\delta(J + \lambda\Lambda)[y_*; \eta_1] = 0$ for all η_1 : y_* is an unconstrained stationary point of $\int(F + \lambda G) dx$, and the Euler–Lagrange equation of $F + \lambda G$ follows by Theorem 2.1. $\square$

E Simultaneous Diagonalization: Proof

Setting (Theorem 5.2): M symmetric positive definite, K symmetric, both real $n \times n$; the problem $K\vec{v} = \omega^2 M\vec{v}$.

Proof. Step 1: the square root. By the spectral theorem [stated; the spectral theorem of linear algebra], $M = O^T D O$ with O orthogonal and $D = \text{diag}(d_1, \dots, d_n)$, $d_i > 0$ (positive definiteness). Define $M^{1/2} \equiv O^T D^{1/2} O$: symmetric, positive definite, invertible, and $(M^{1/2})^2 = M$.

Step 2: reduce to an ordinary symmetric problem. Substitute $\vec{w} = M^{1/2}\vec{v}$ in the generalized problem:

$$K M^{-1/2} \vec{w} = \omega^2 M^{1/2} \vec{w} \iff \underbrace{M^{-1/2} K M^{-1/2}}_{\equiv A} \vec{w} = \omega^2 \vec{w}, \quad (236)$$

and A is symmetric ($A^T = M^{-1/2} K^T M^{-1/2} = A$ since $M^{-1/2}$, K are). The spectral theorem, invoked once more, gives an orthonormal basis $\vec{w}_1, \dots, \vec{w}_n$ of eigenvectors with real eigenvalues ω_r^2 .

Step 3: translate back. Set $\vec{v}_r = M^{-1/2} \vec{w}_r$: a basis (image of a basis under an invertible map), solving the generalized problem, with the claimed orthogonality

$$\vec{v}_r^T M \vec{v}_s = \vec{w}_r^T M^{-1/2} M M^{-1/2} \vec{w}_s = \vec{w}_r^T \vec{w}_s = \delta_{rs}. \quad (237)$$

If K is positive definite, the eigenvalues are positive: $\omega_r^2 = \vec{w}_r^T A \vec{w}_r = (M^{-1/2} \vec{w}_r)^T K (M^{-1/2} \vec{w}_r) = \vec{v}_r^T K \vec{v}_r > 0$.

Step 4: decoupling. Expand $\vec{\theta}(t) = \sum_r \xi_r(t) \vec{v}_r$ and insert into $M\ddot{\vec{\theta}} = -K\vec{\theta}$; multiply on the left by $\vec{v}_s^T$. Using $K\vec{v}_r = \omega_r^2 M\vec{v}_r$ and then M -orthonormality,

$$\sum_r \ddot{\xi}_r \vec{v}_s^T M \vec{v}_r = - \sum_r \xi_r \omega_r^2 \vec{v}_s^T M \vec{v}_r \implies \ddot{\xi}_s = -\omega_s^2 \xi_s \quad (s = 1, \dots, n). \quad (238) \quad \square$$

F The Jacobi Identity: Computation

We verify $\mathcal{J} \equiv \{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$ for one degree of freedom; the n -degree case is the identical computation repeated for each index pair, since the bracket is a sum over i of identical one-pair blocks. Write partials as subscripts ($f_q = \partial f / \partial q$, $f_{qp} = \partial^2 f / \partial q \partial p$, etc.), and recall $f_{qp} = f_{pq}$ throughout (C^2).

The organizing observation. Each double bracket expands into terms containing exactly *one* second derivative (of the innermost functions) times two first derivatives. Sort all terms of $\mathcal{J}$ into three groups by which function carries the second derivative; we show each group cancels on its own. By the cyclic symmetry of $\mathcal{J}$, it suffices to check the group of second derivatives of f — the other two follow by relabeling.

The expansion. From $\{g, h\} = g_q h_p - g_p h_q$:

$$\begin{aligned} \{f, \{g, h\}\} &= f_q \partial_p (g_q h_p - g_p h_q) - f_p \partial_q (g_q h_p - g_p h_q) \\ &= f_q (g_{qp} h_p + g_q h_{pp} - g_{pp} h_q - g_p h_{qp}) - f_p (g_{qq} h_p + g_q h_{pq} - g_{pq} h_q - g_p h_{qq}). \end{aligned} \quad (239)$$

No second derivative of f appears here — f sits outside. Second derivatives of f arise only from the two brackets in which f is inside:

$$\{g, \{h, f\}\} \Big|_{f''\text{-terms}} = g_q (h_q f_{pp} - h_p f_{qp}) - g_p (h_q f_{pq} - h_p f_{qq}), \quad (240)$$

$$\{h, \{f, g\}\} \Big|_{f''\text{-terms}} = h_q (f_{qp} g_p - f_{pp} g_q) - h_p (f_{qq} g_p - f_{pq} g_q). \quad (241)$$

Add and collect by second-derivative type:

$$\begin{aligned} f_{pp} : \quad & g_q h_q - h_q g_q = 0, \\ f_{qq} : \quad & g_p h_p - h_p g_p = 0, \\ f_{qp} = f_{pq} : \quad & (-g_q h_p) + (-g_p h_q) + (h_q g_p) + (h_p g_q) = 0. \end{aligned} \quad (242)$$

Every coefficient vanishes — crucially using $f_{qp} = f_{pq}$ to merge the mixed terms; the Jacobi identity is, at bottom, *the symmetry of second derivatives*, three times over. The g'' - and h'' -groups vanish by the cyclic relabelings $(f, g, h) \rightarrow (g, h, f) \rightarrow (h, f, g)$, and with all second-derivative groups zero and no other terms present, $\mathcal{J} = 0$.